Middleware

Assignment 1:

Install and Configure Middleware

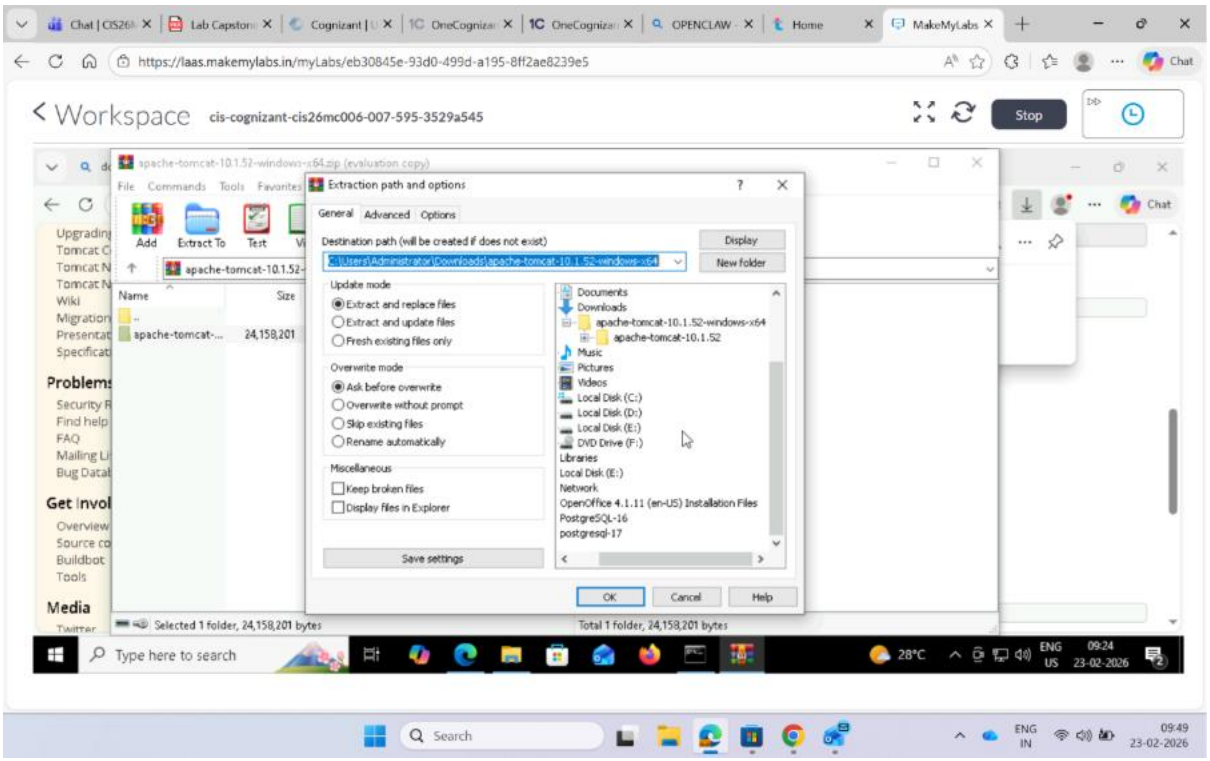
As part of this assignment, you need to complete below tasks:

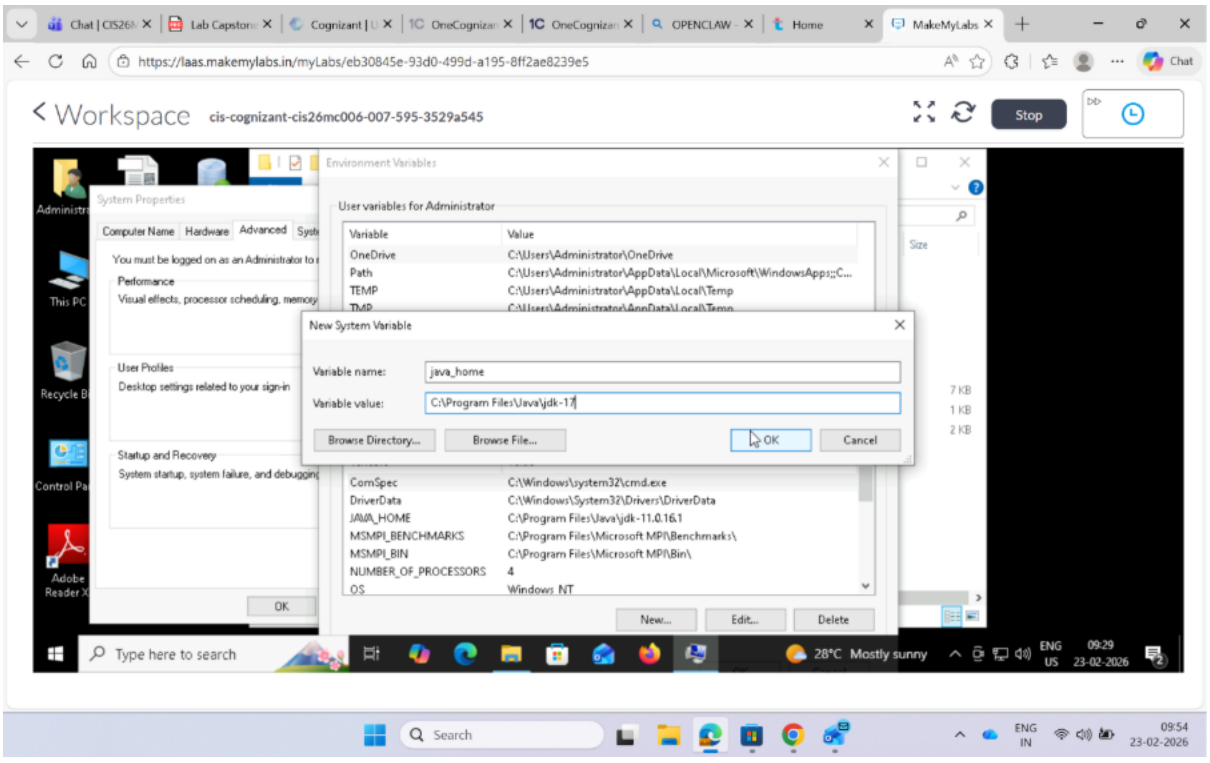
1. Download and install Apache Tomcat

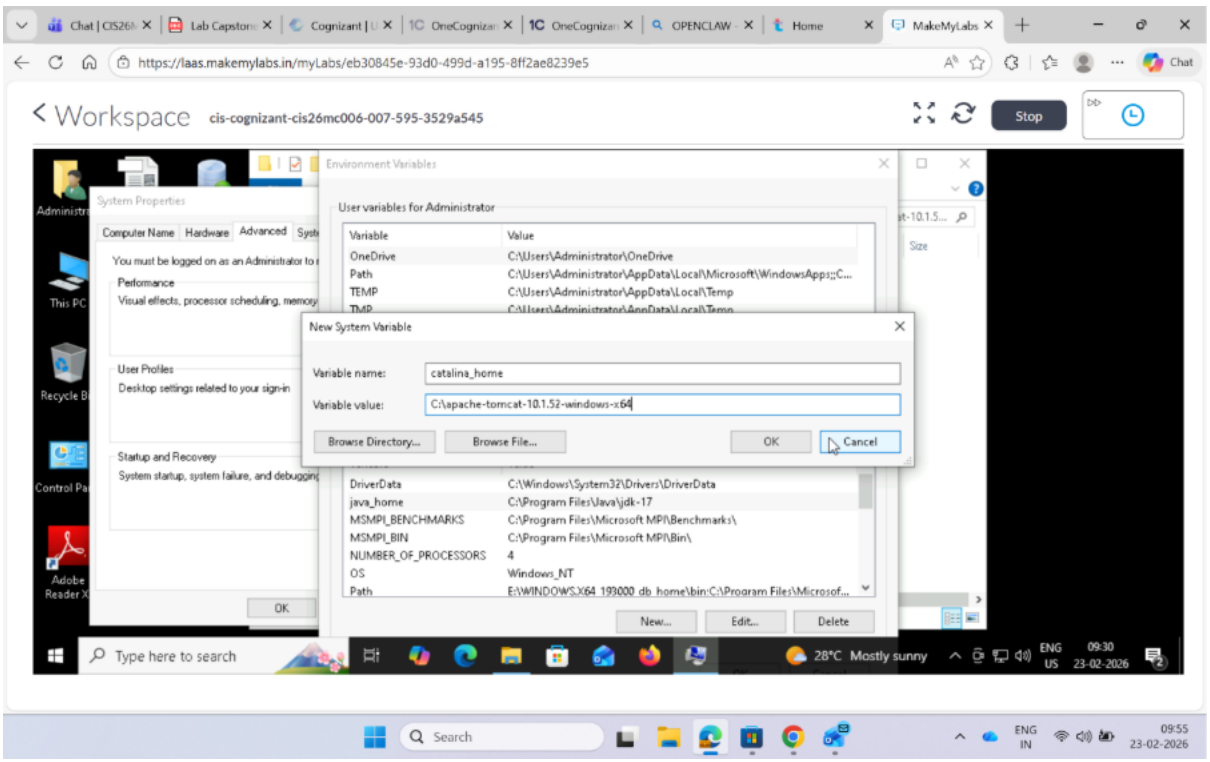
2. Set up a basic Java web application

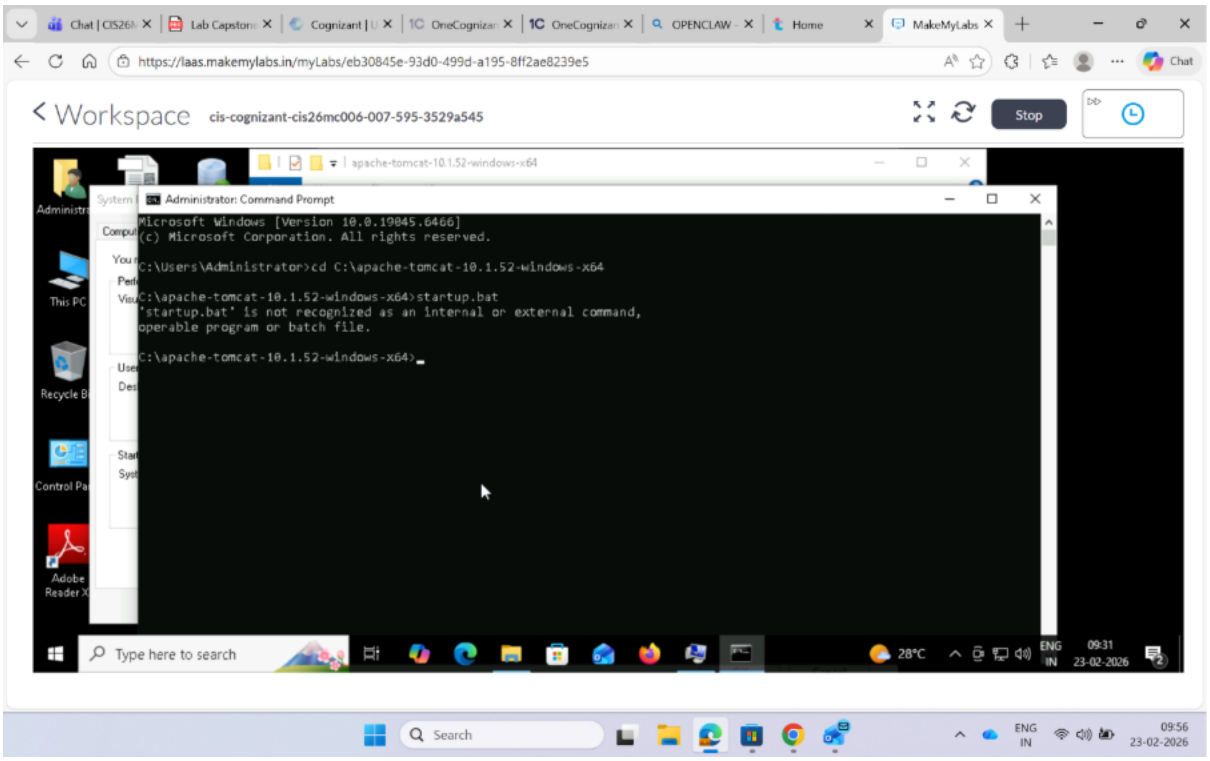
3. Configure environment variables

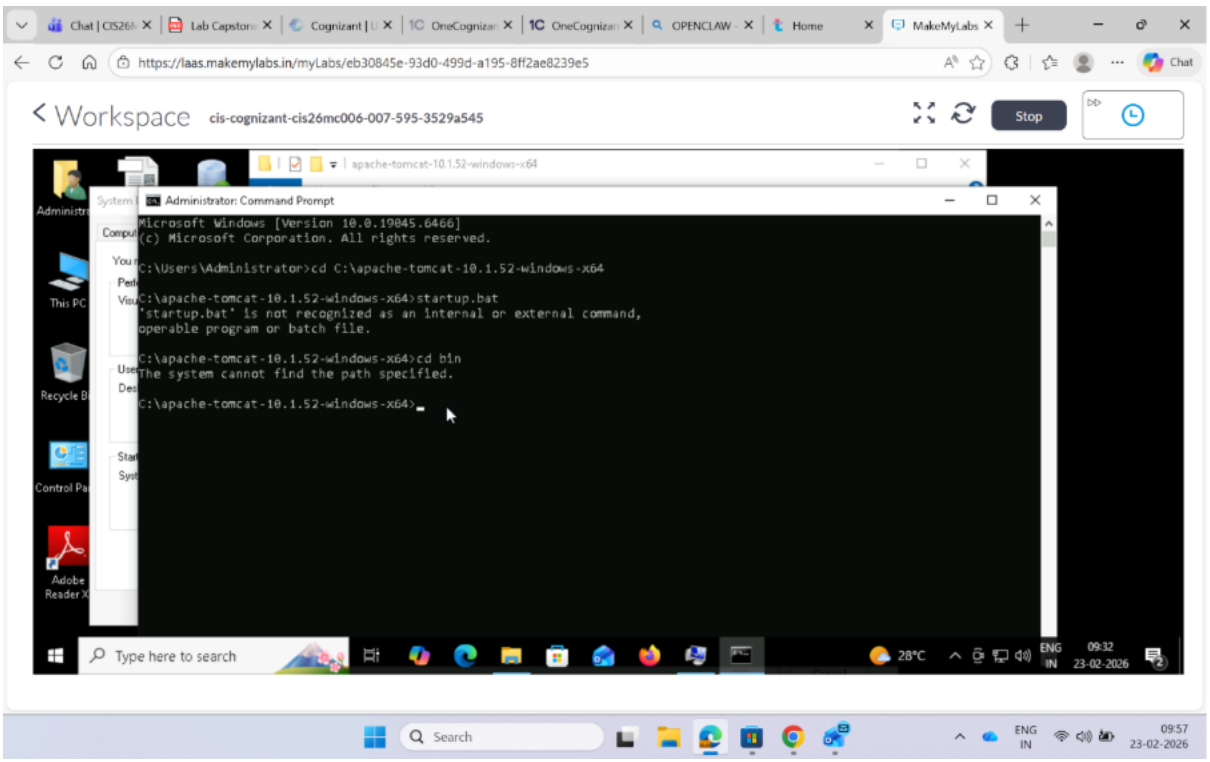
4. Test server connectivity and deployment

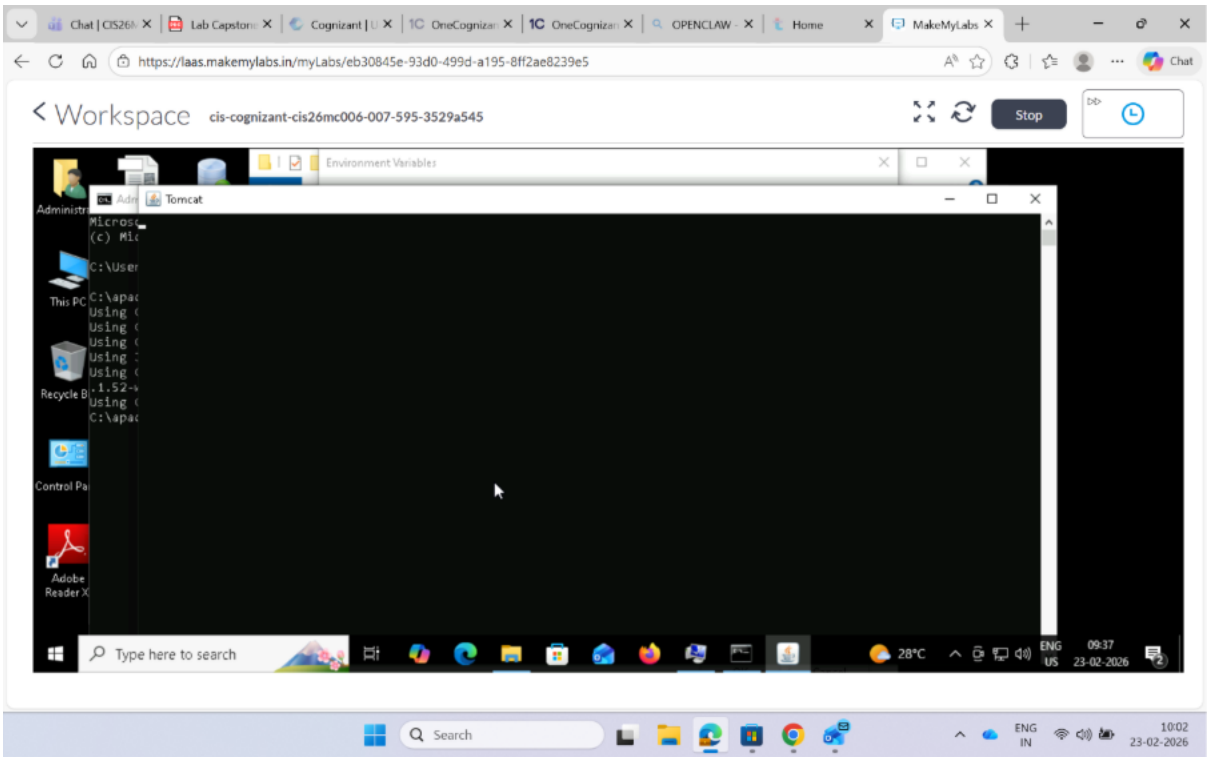


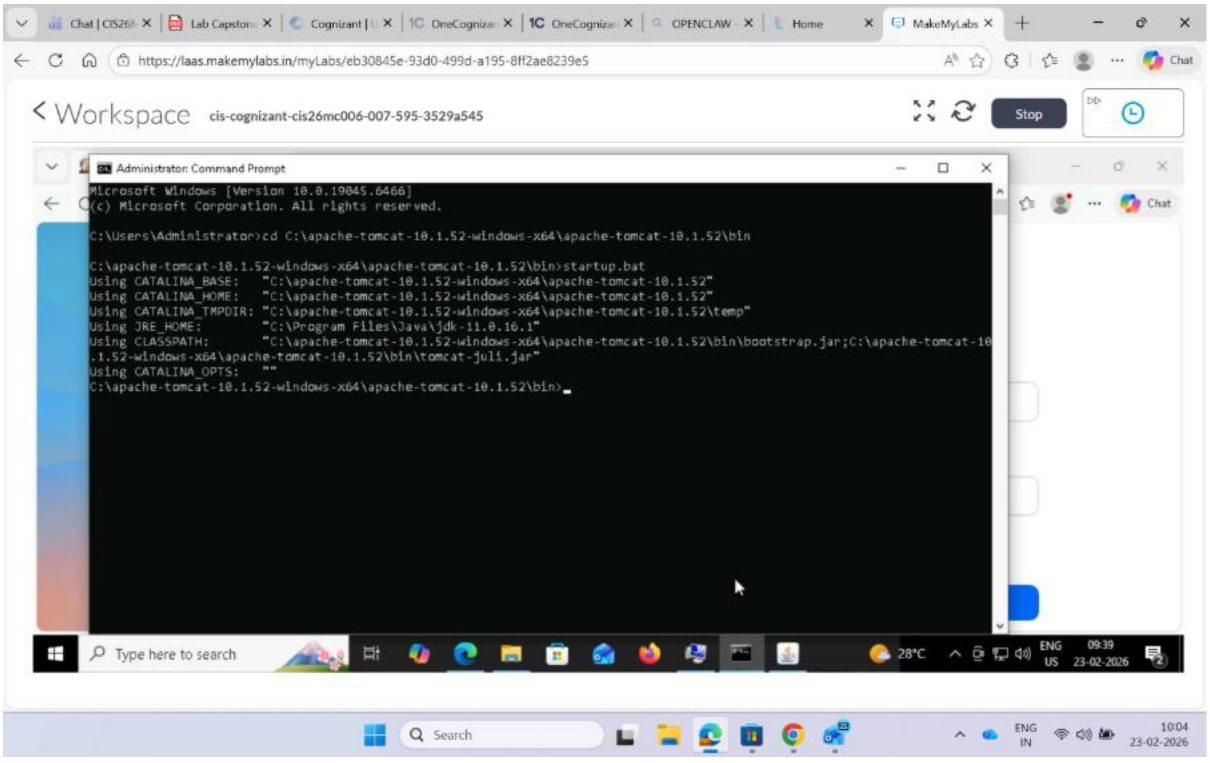

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Environment Variables
User variables for Administrator
Variable
Value
OneDrive
C:\Users\Administrator\OneDrive
Path
C:\Users\Administrator\AppData\Local\Microsoft\WindowsApps;;C...
TEMP
C:\Users\Administrator\AppData\Local\Temp
New System Variable
Variable name:
java_home
Variable value:
C:\Program Files\Java\jdk-17
Browse Directory...
Browse File...
OK
Cancel
ComSpec
C:\Windows\system32\cmd.exe
DriverData
C:\Windows\System32\Drivers\DriverData
JAVA_HOME
C:\Program Files\Java\jdk-11.0.16.1
MSMPI_BENCHMARKS
C:\Program Files\Microsoft MPI\Benchmarks\
MSMPI_BIN
C:\Program Files\Microsoft MPI\Bin\
NUMBER_OF_PROCESSORS
4
OS
Windows_NT
New...
Edit...
Delete
System Properties
Computer Name
Hardware
Advanced
Performance
User Profiles
Startup and Recovery
Type here to search
28°C Mostly sunny
ENG
US
09:29
23-02-2026
Search
ENG
IN
09:54
23-02-2026

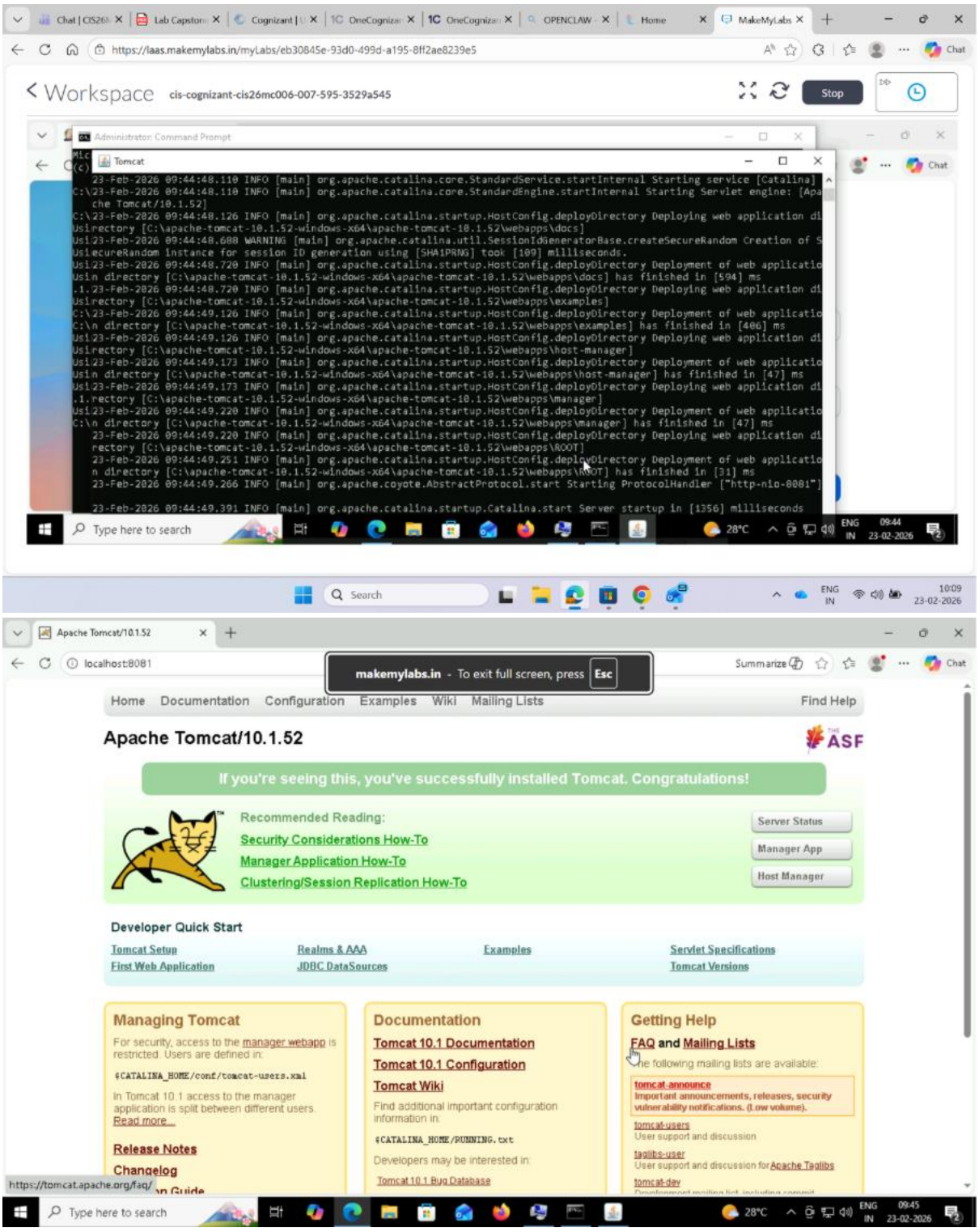

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Environment Variables
User variables for Administrator
Variable
Value
OneDrive
C:\Users\Administrator\OneDrive
Path
C:\Users\Administrator\AppData\Local\Microsoft\WindowsApps;;C...
TEMP
C:\Users\Administrator\AppData\Local\Temp
New System Variable
Variable name:
catalina_home
Variable value:
C:\apache-tomcat-10.1.52-windows-x64
Browse Directory...
Browse File...
OK
Cancel
DriverData
C:\Windows\System32\Drivers\DriverData
java_home
C:\Program Files\Java\jdk-17
MSMPI_BENCHMARKS
C:\Program Files\Microsoft MPI\Benchmarks\
MSMPI_BIN
C:\Program Files\Microsoft MPI\Bin\
NUMBER_OF_PROCESSORS
4
OS
Windows_NT
Path
New...
Edit...
Delete
System Properties
Computer Name
Hardware
Advanced
Performance
User Profiles
Startup and Recovery
Type here to search
28°C Mostly sunny
ENG
US
09:30
23-02-2026
Search
ENG
IN
09:55
23-02-2026

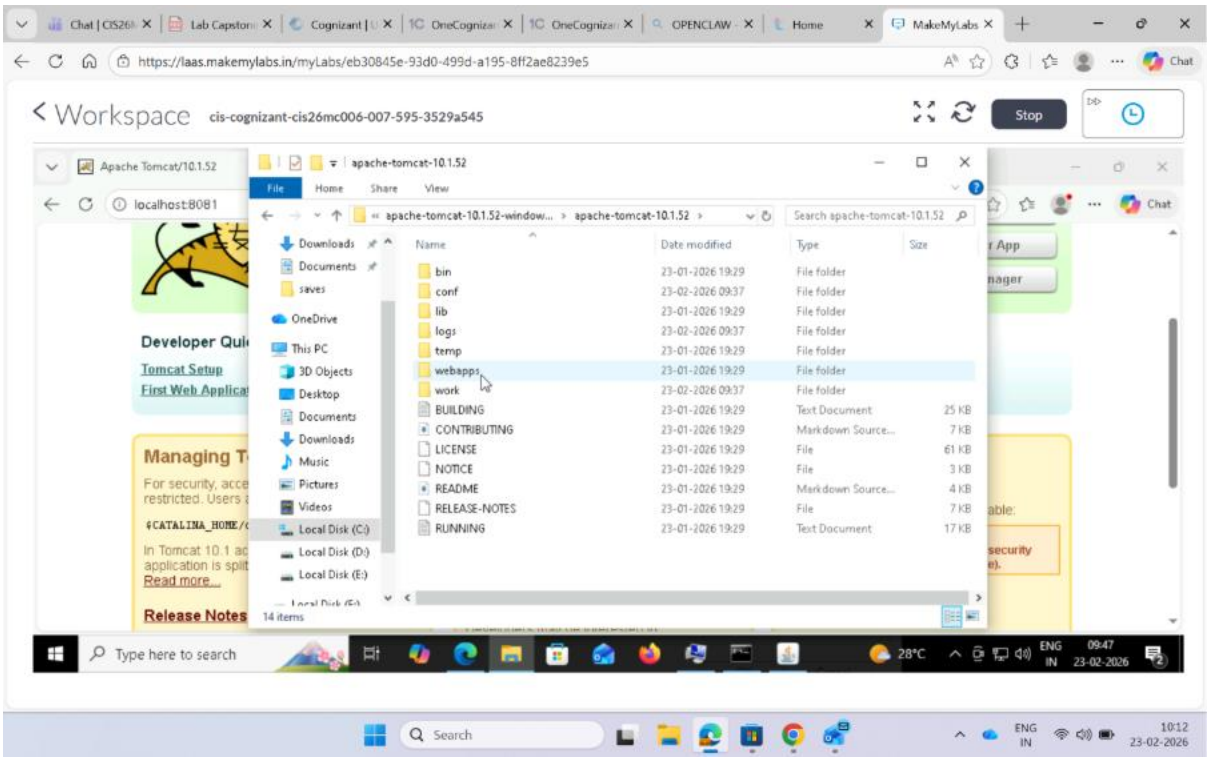

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\apache-tomcat-10.1.52-windows-x64
C:\apache-tomcat-10.1.52-windows-x64>startup.bat
'startup.bat' is not recognized as an internal or external command,
operable program or batch file.
C:\apache-tomcat-10.1.52-windows-x64>
Type here to search
28°C
ENG
IN
09:31
23-02-2026

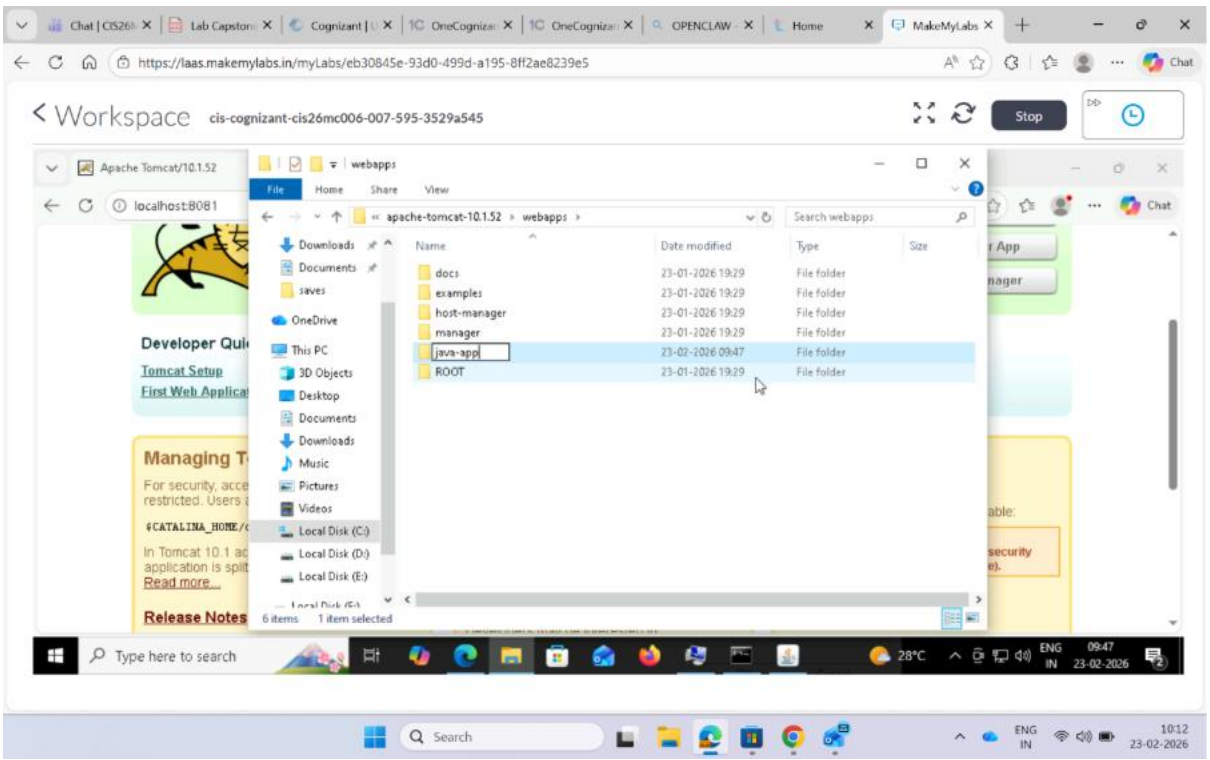

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\apache-tomcat-10.1.52-windows-x64
C:\apache-tomcat-10.1.52-windows-x64>startup.bat
'startup.bat' is not recognized as an internal or external command,
operable program or batch file.
C:\apache-tomcat-10.1.52-windows-x64>cd bin
The system cannot find the path specified.
C:\apache-tomcat-10.1.52-windows-x64>
Type here to search
28°C
ENG
IN
09:32
23-02-2026

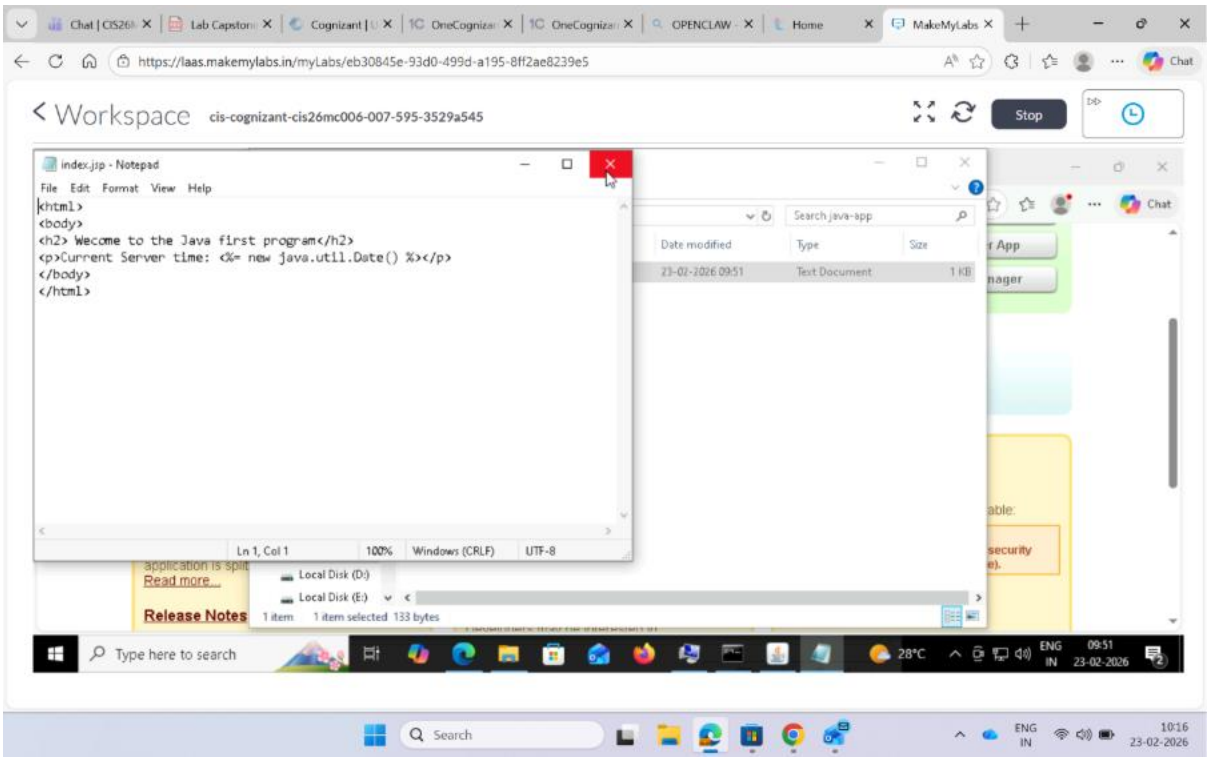

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Environment Variables
Tomcat
Type here to search
28°C
ENG US
09:37
23-02-2026
Search
ENG IN
10:02
23-02-2026

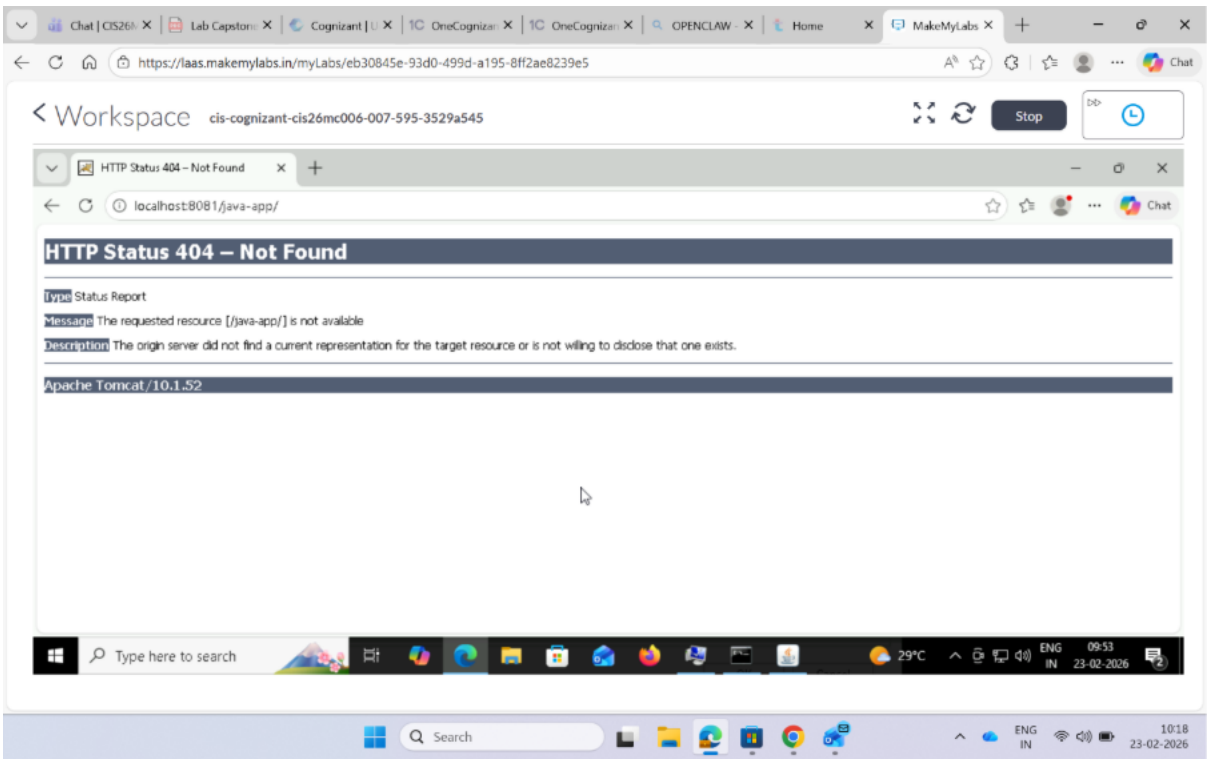

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52\bin
C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52\bin>startup.bat
Using CATALINA_BASE: "C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52"
Using CATALINA_HOME: "C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52"
Using CATALINA_TMPDIR: "C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52\temp"
Using JRE_HOME: "C:\Program Files\Java\jdk-11.0.16.1"
Using CLASSPATH: "C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52\bin\bootstrap.jar;C:\apache-tomcat-10
.1.52-windows-x64\apache-tomcat-10.1.52\bin\tomcat-juli.jar"
Using CATALINA_OPTS: ""
C:\apache-tomcat-10.1.52-windows-x64\apache-tomcat-10.1.52\bin>
Type here to search
28°C
ENG US
09:39
23-02-2026
Search
ENG IN
10:04
23-02-2026

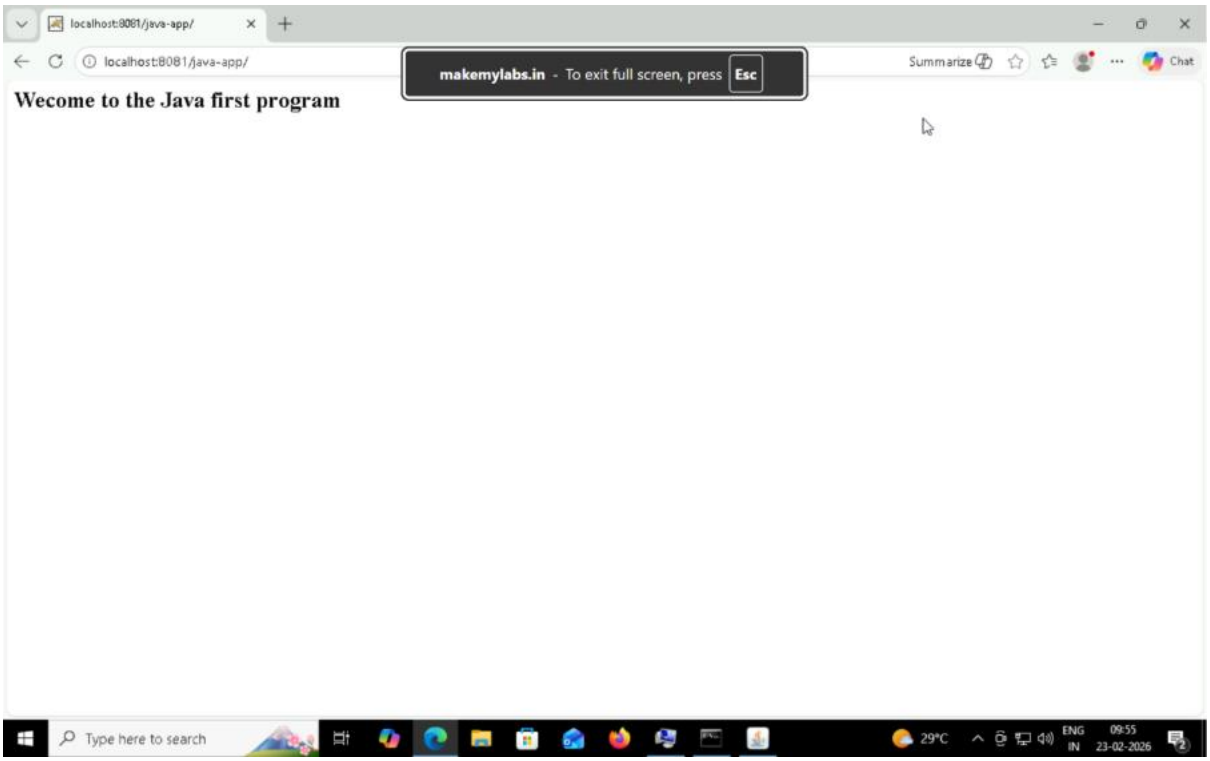

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Tomcat
Apache Tomcat/10.1.52
If you're seeing this, you've successfully installed Tomcat. Congratulations!
Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To
Server Status
Manager App
Host Manager
Developer Quick Start
Tomcat Setup
First Web Application
Realms & AAA
JDBC DataSources
Examples
Servlet Specifications
Tomcat Versions
Managing Tomcat
Documentation
Tomcat 10.1 Documentation
Tomcat 10.1 Configuration
Tomcat Wiki
Getting Help
FAQ and Mailing Lists
Release Notes
Changelog

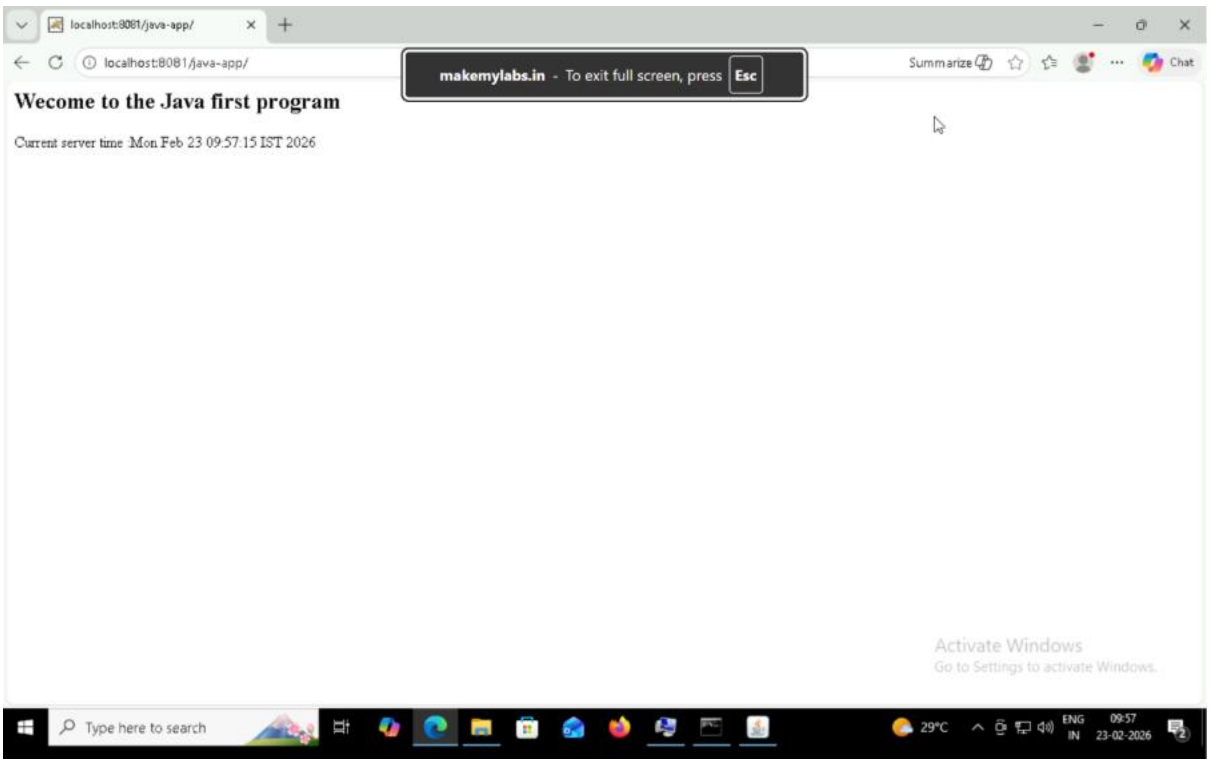

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Apache Tomcat/10.1.52
localhost:8081
apache-tomcat-10.1.52
bin
conf
lib
logs
temp
webapps
work
BUILDING
CONTRIBUTING
LICENSE
NOTICE
README
RELEASE-NOTES
RUNNING
Tomcat Setup
Release Notes
Type here to search


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Apache Tomcat/10.1.52
localhost:8081
webapps
docs
examples
host-manager
manager
java-app
ROOT
Tomcat Setup
Release Notes
Type here to search


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
index.jsp - Notepad
File Edit Format View Help
<html>
<body>
<h2> Wecome to the Java first program</h2>
Current Server time: <%= new java.util.Date() %>
</body>
</html>
Ln 1, Col 1
100%
Windows (CRLF)
UTF-8
Search java-app
Date modified
Type
Size
23-02-2026 09:51
Text Document
1 KB
Local Disk (D:)
Local Disk (E:)
1 item
1 item selected 133 bytes
Release Notes
Type here to search
28°C
ENG
IN
09:51
23-02-2026


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
HTTP Status 404 – Not Found
localhost:8081/java-app/
HTTP Status 404 – Not Found
Type Status Report
Message The requested resource [/java-app/] is not available
Description The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.
Apache Tomcat/10.1.52
Type here to search
29°C
ENG
IN
09:53
23-02-2026


localhost:8081/java-app/
localhost:8081/java-app/
makemylabs.in - To exit full screen, press Esc
Summarize
Chat
Wecome to the Java first program
Type here to search
29°C
ENG
IN
09:55
23-02-2026


localhost:8081/java-app/
localhost:8081/java-app/
makemylabs.in - To exit full screen, press Esc
Summarize
Chat
Wecome to the Java first program
Current server time :Mon Feb 23 09:57:15 IST 2026
Activate Windows
Go to Settings to activate Windows.
Type here to search
29°C
ENG
IN
09:57
23-02-2026

Assignment 2:

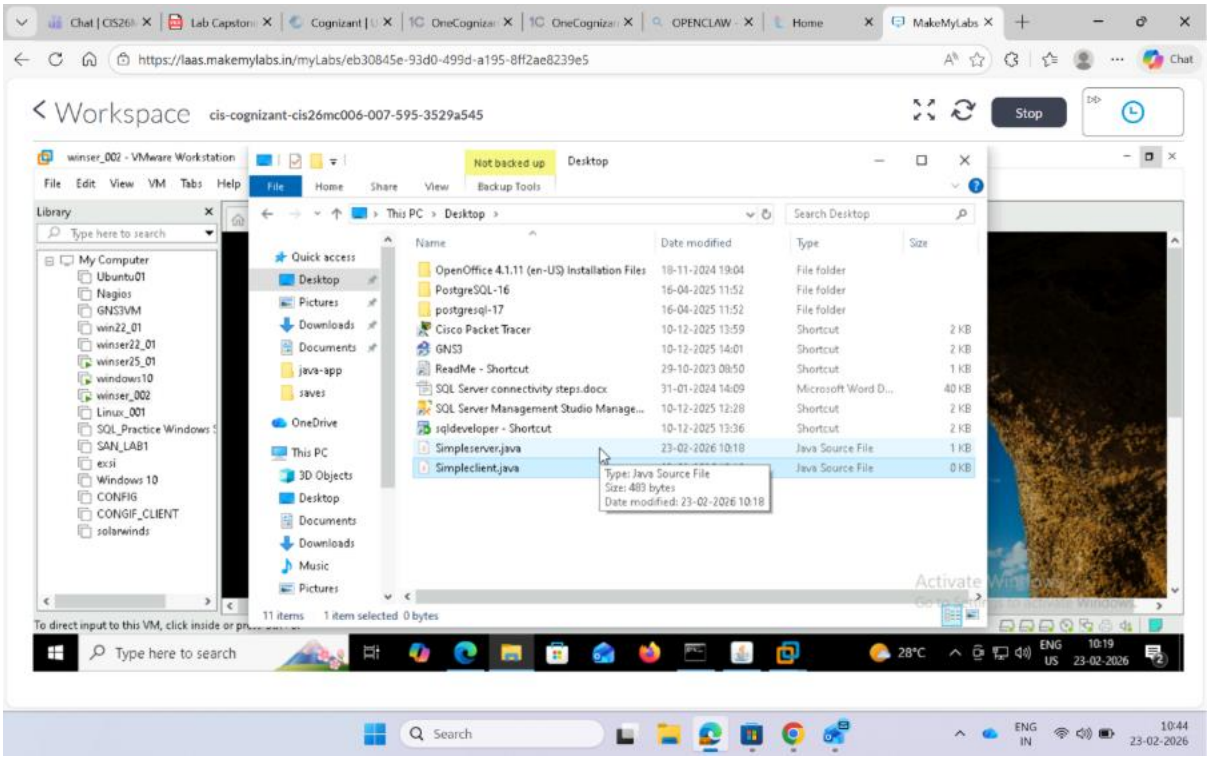
Middleware Communication Simulation

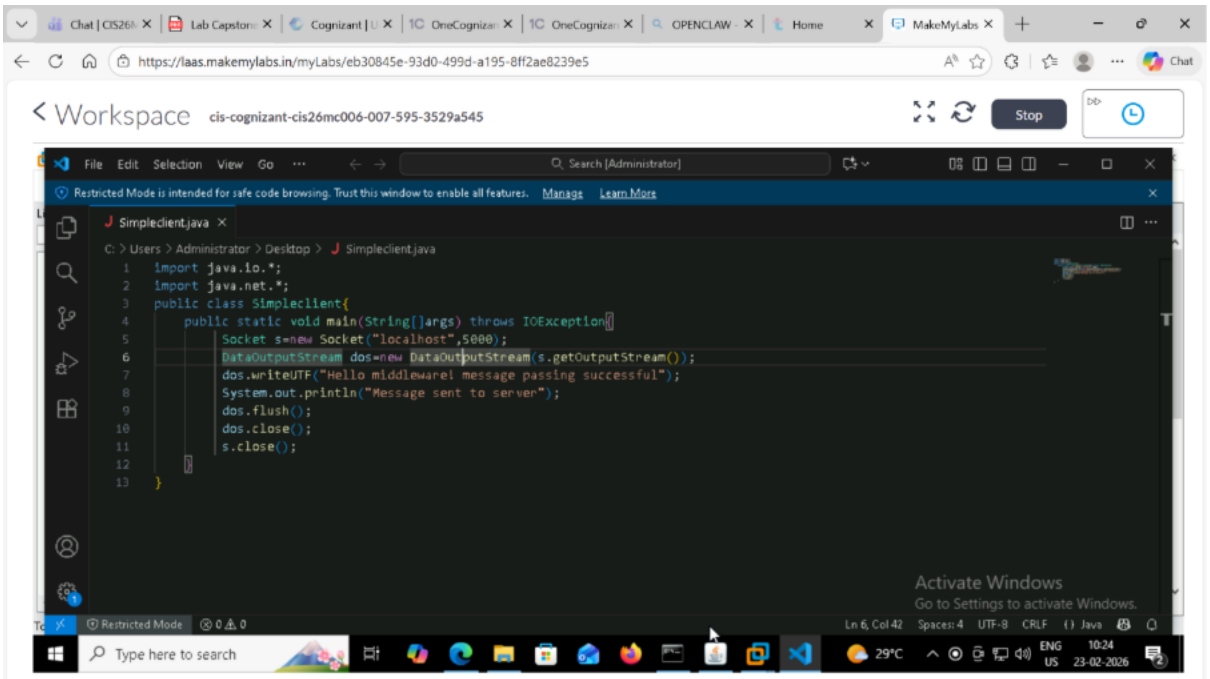
As part of this assignment, you need to complete below tasks:

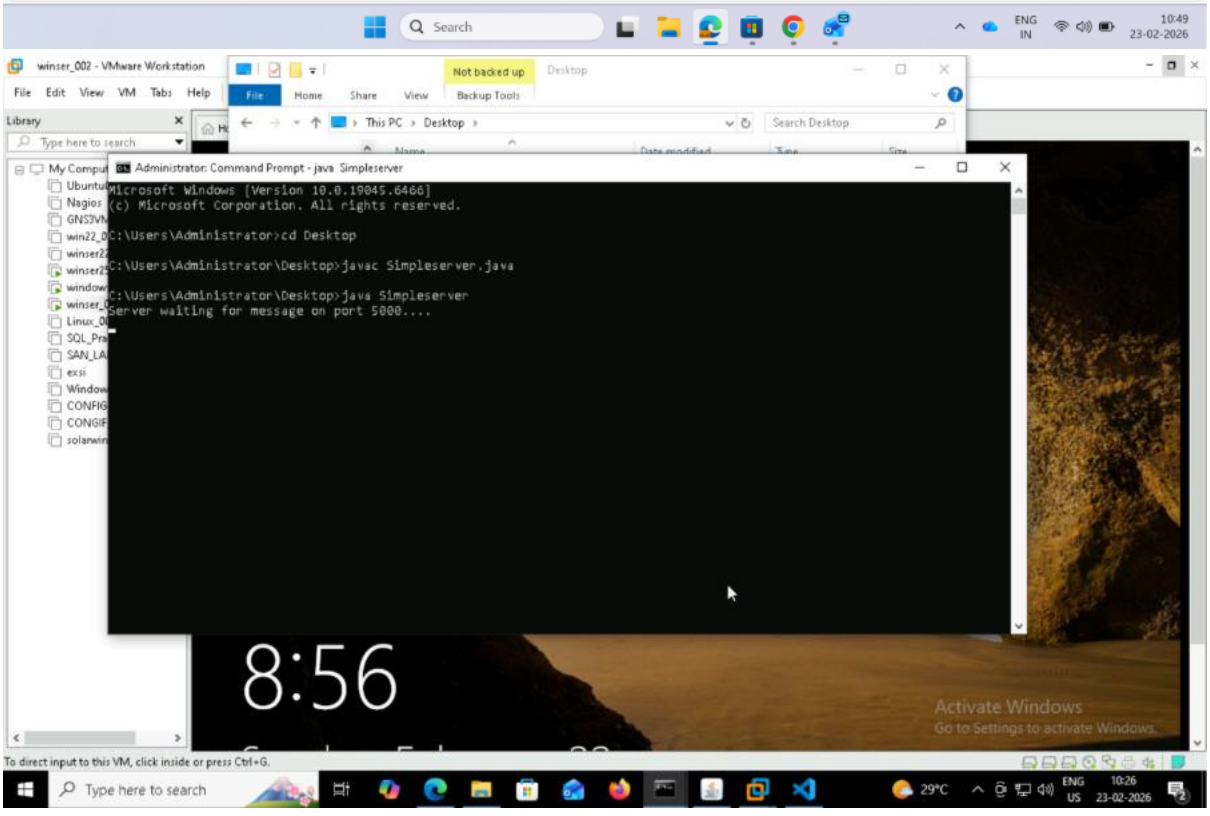
1.Create a simple client-server application demonstrating message passing

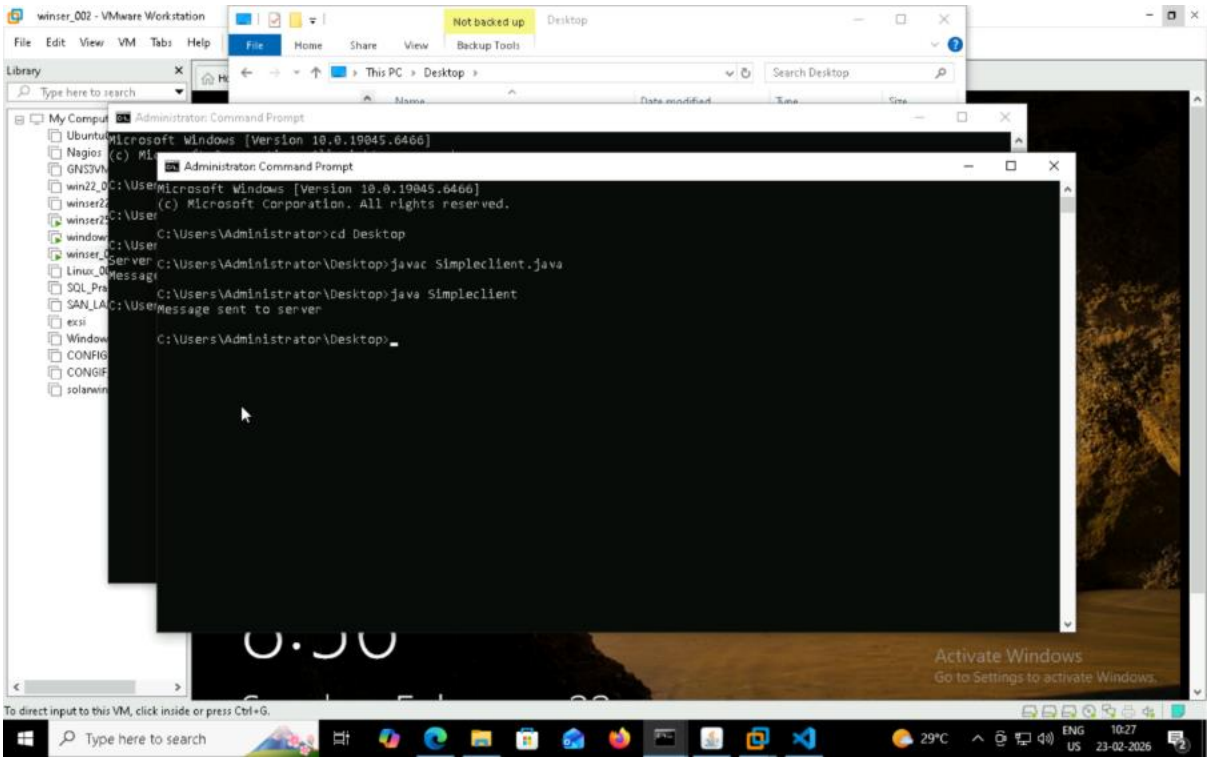
2. Implement a basic middleware communication protocol

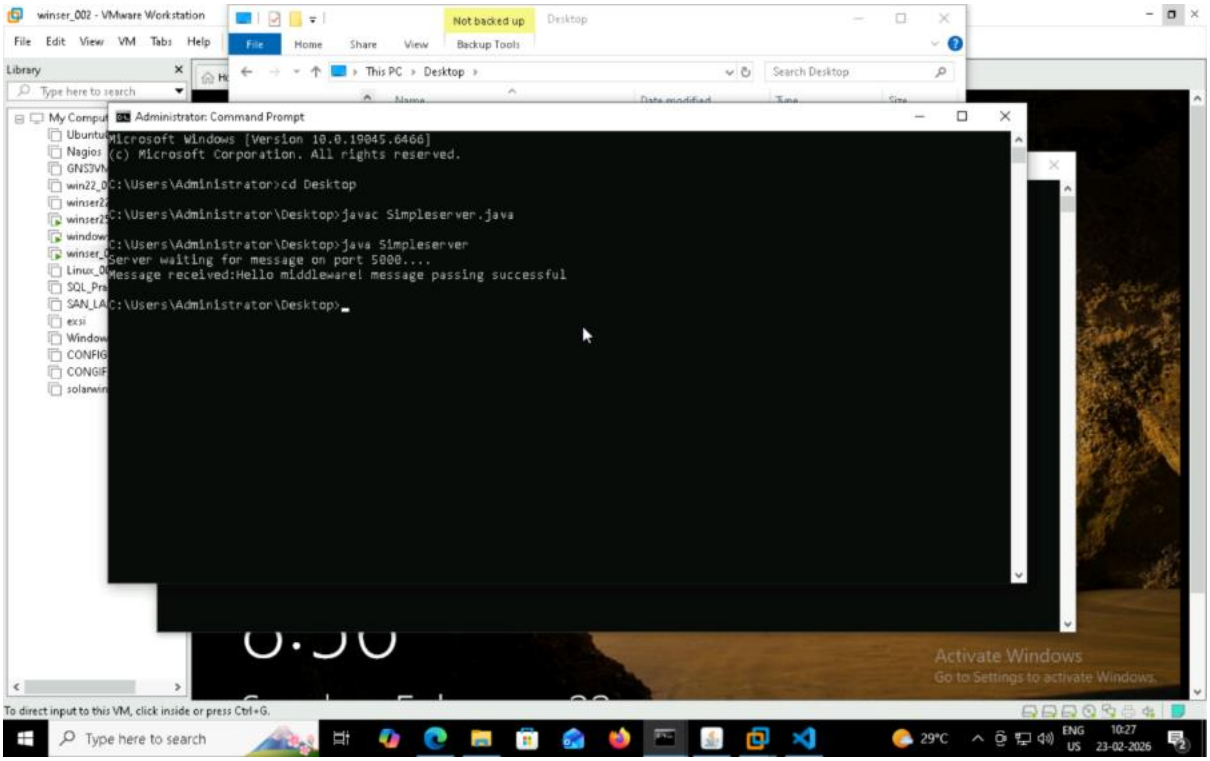
3.Develop a small application showing inter-process communication




Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Simpleclient.java
import java.io.*;
import java.net.*;
public class Simpleclient{
public static void main(String[]args) throws IOException{
Socket s=new Socket("localhost",5000);
DataOutputStream dos=new DataOutputStream(s.getOutputStream());
dos.writeUTF("Hello middleware! message passing successful");
System.out.println("Message sent to server");
dos.flush();
dos.close();
s.close();
}
}
Activate Windows
Go to Settings to activate Windows.


winser_002 - VMware Workstation
Administrator: Command Prompt - java Simpleserver
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd Desktop
C:\Users\Administrator\Desktop>javac Simpleserver.java
C:\Users\Administrator\Desktop>java Simpleserver
Server waiting for message on port 5000....
8:56
Activate Windows
Go to Settings to activate Windows.


winser_002 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
Not backed up
Desktop
File Home Share View Backup Tools
This PC › Desktop ›
Search Desktop
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd Desktop
C:\Users\Administrator\Desktop>javac Simpleclient.java
C:\Users\Administrator\Desktop>java Simpleclient
Message sent to server
C:\Users\Administrator\Desktop>
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
29°C
ENG
US
10:27
23-02-2026


winser_002 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
Not backed up
Desktop
File Home Share View Backup Tools
This PC › Desktop ›
Search Desktop
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd Desktop
C:\Users\Administrator\Desktop>javac Simpleserver.java
C:\Users\Administrator\Desktop>java Simpleserver
Server waiting for message on port 5000....
Message received:Hello middleware! message passing successful
C:\Users\Administrator\Desktop>
Activate Windows
Go to Settings to activate Windows.
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
29°C
ENG
US
10:27
23-02-2026

Assignment 3:

Web Server Setup and Configuration As part of this assignment, you need to complete below tasks:

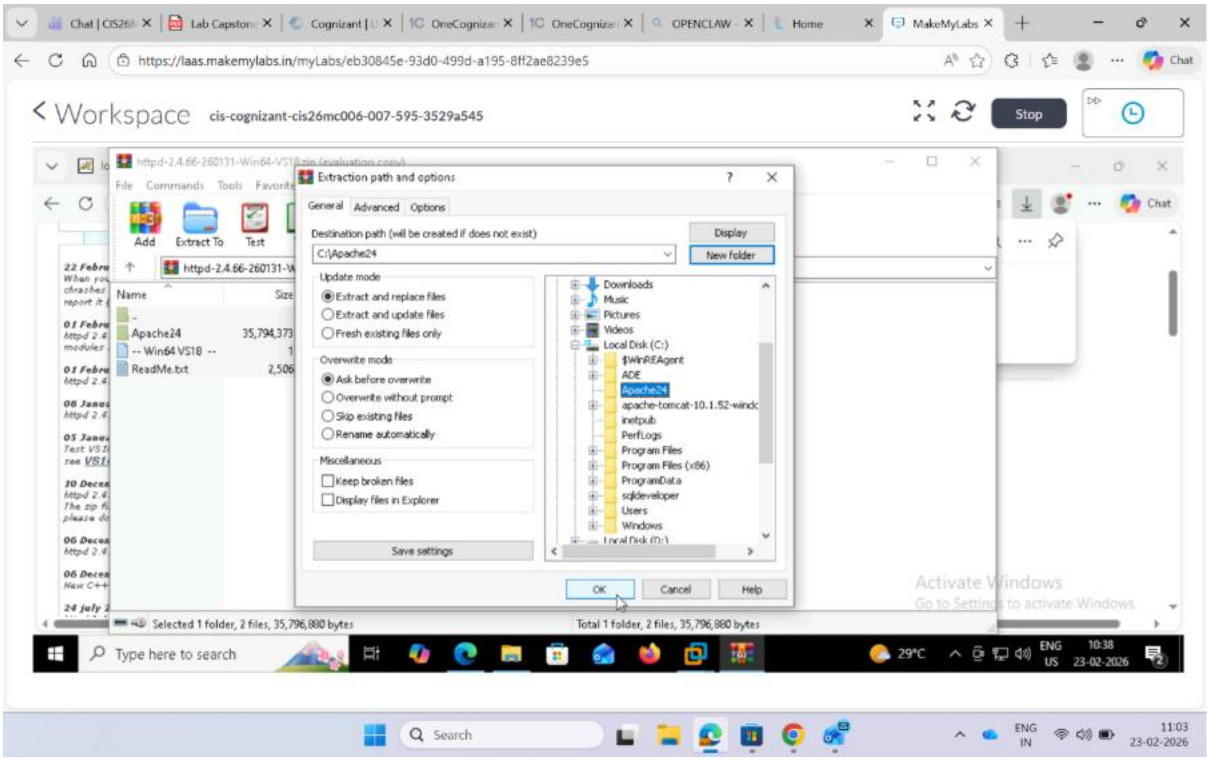
1. Install Apache HTTP Server

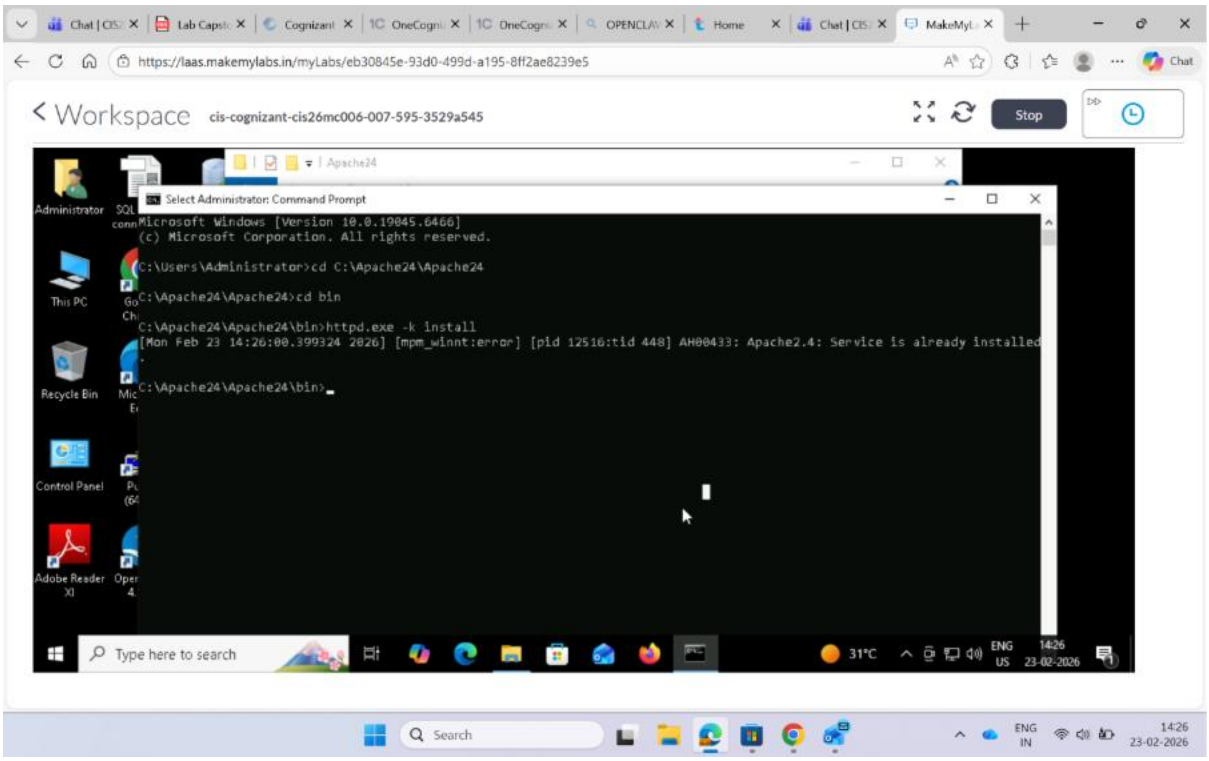
2. Configure virtual hosts

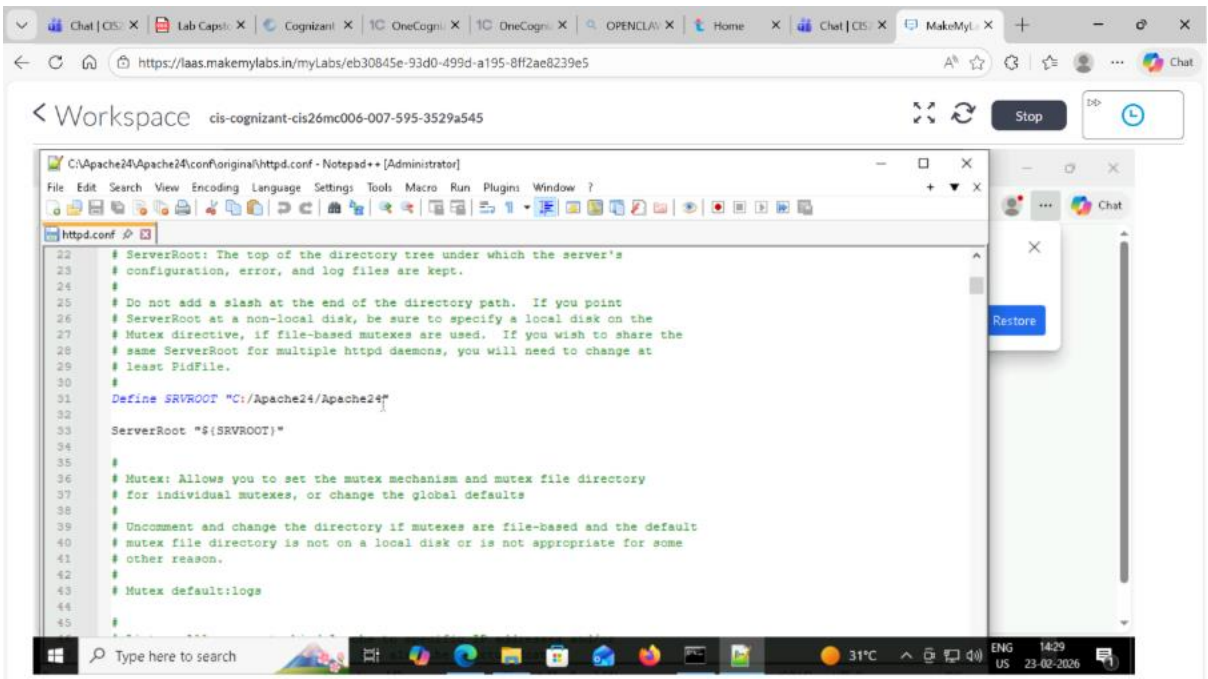
3. Set up basic routing rules

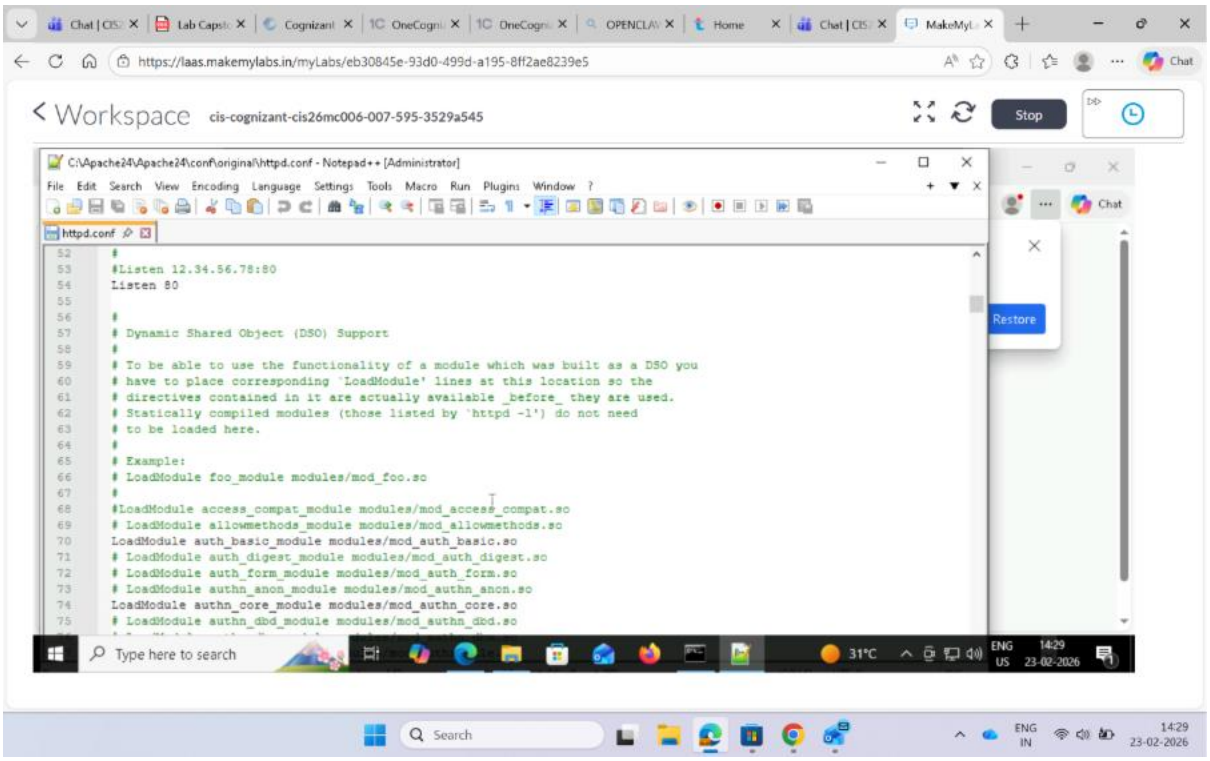
4. Implement simple load balancing

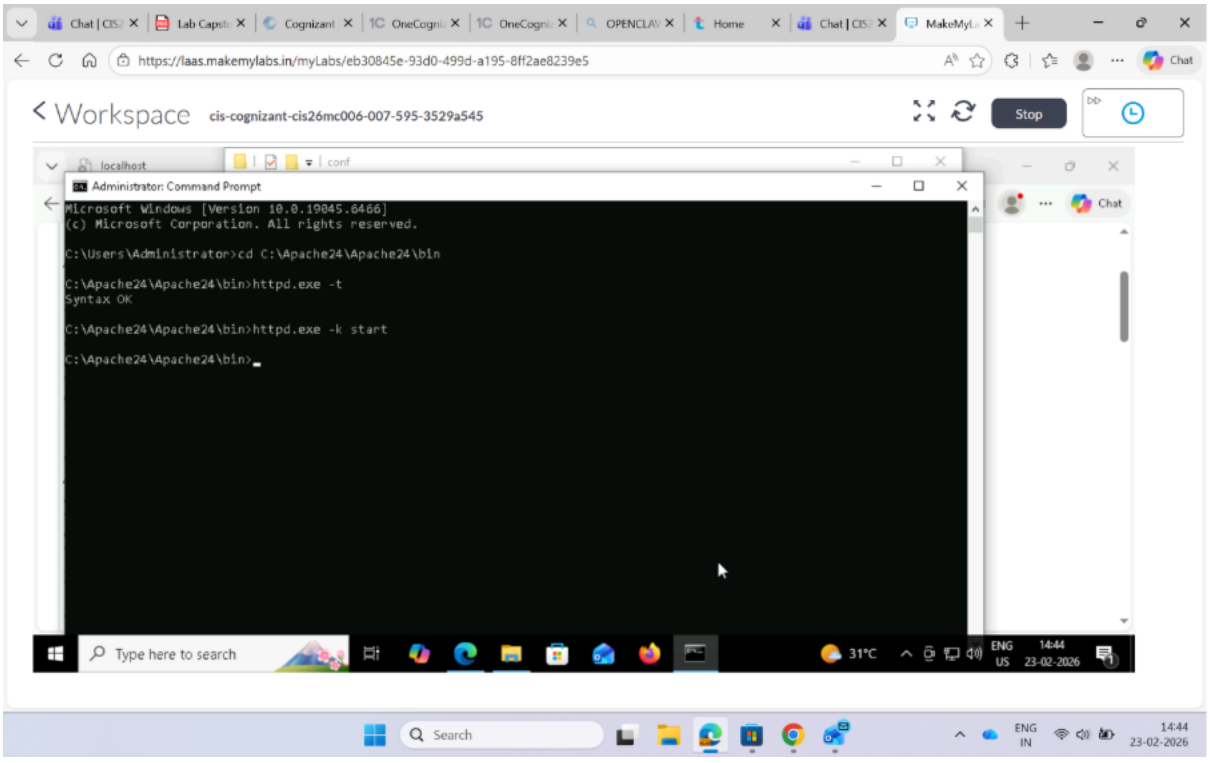
5. Create and deploy a static website

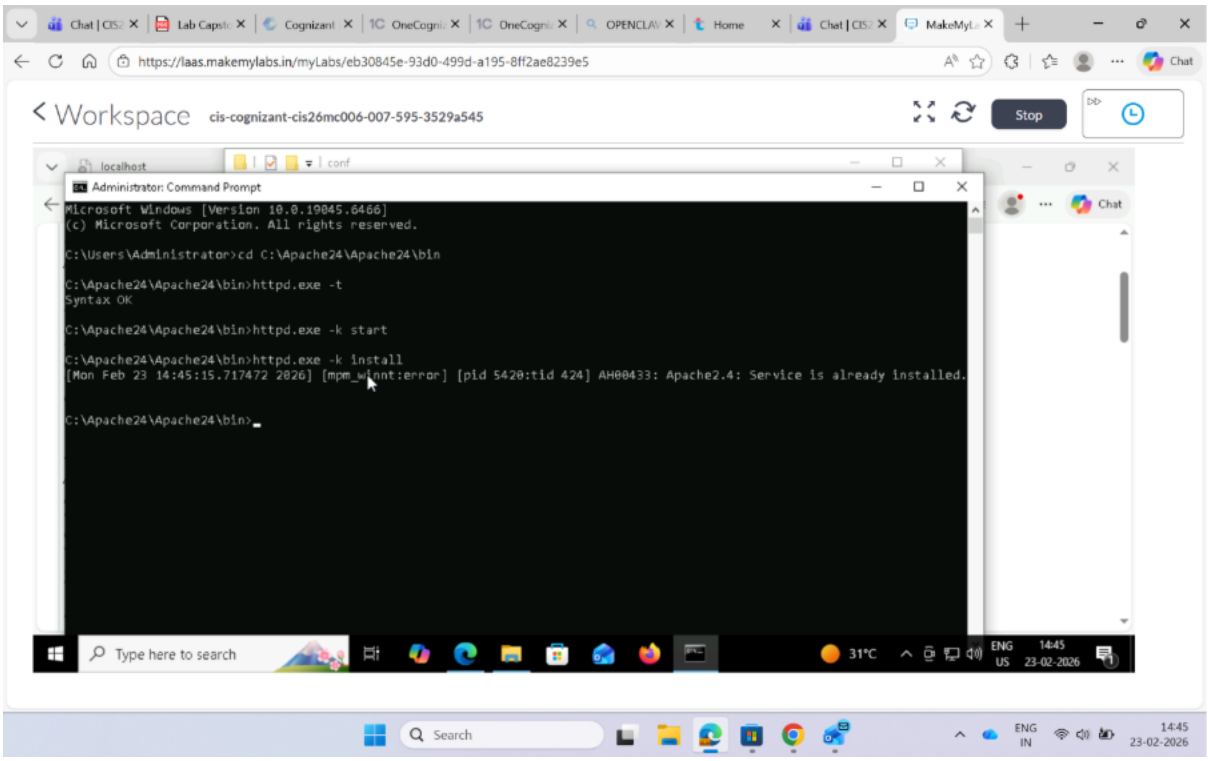


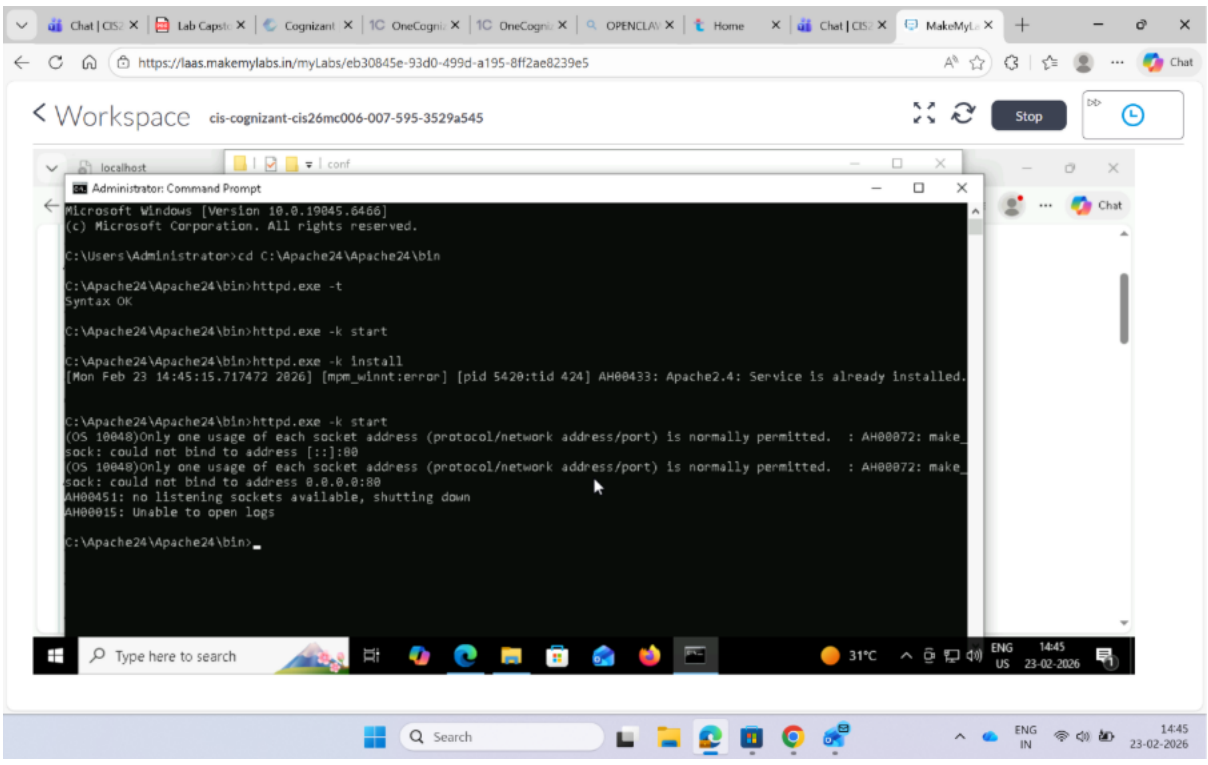

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Select Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\Apache24\Apache24
C:\Apache24\Apache24>cd bin
C:\Apache24\Apache24\bin>httpd.exe -k install
[Mon Feb 23 14:26:08.399324 2026] [mpm_winnt:error] [pid 12516:tid 448] AH00433: Apache2.4: Service is already installed
C:\Apache24\Apache24\bin>
Type here to search
31°C
ENG
US
14:26
23-02-2026

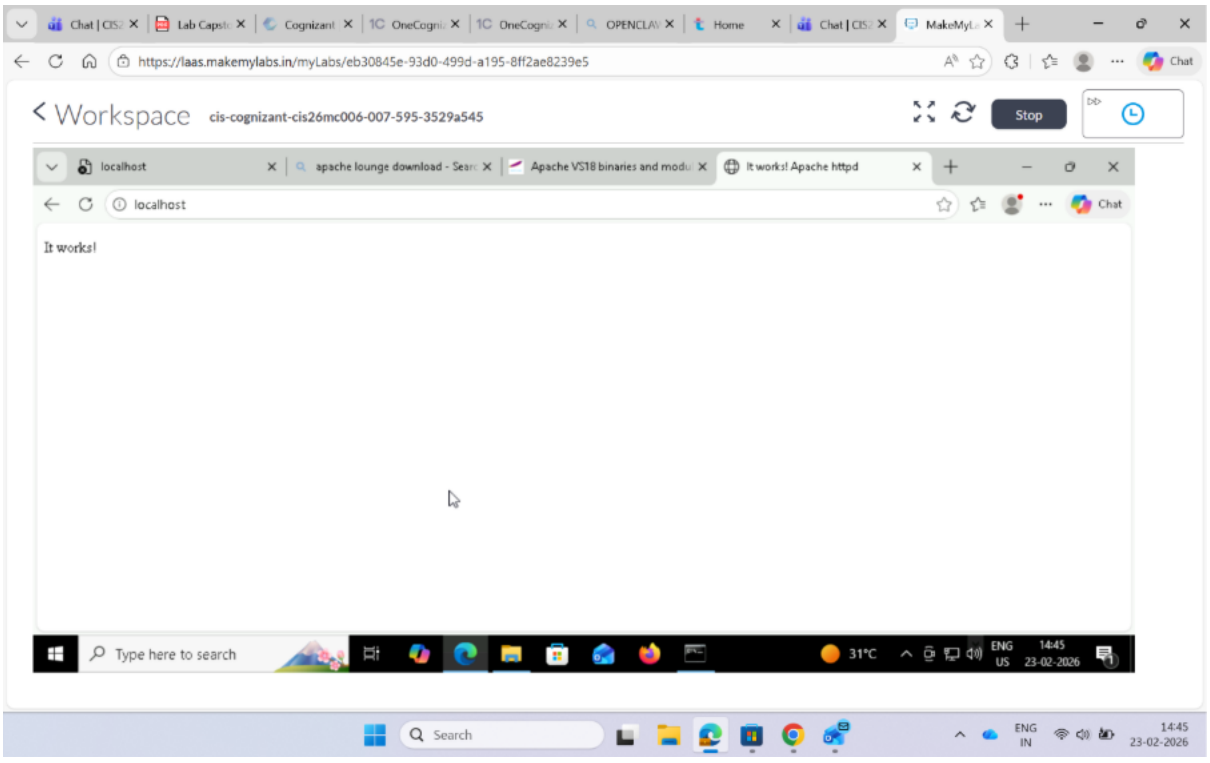

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
C:\Apache24\Apache24\conf\original\httpd.conf - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
httpd.conf
ServerRoot: The top of the directory tree under which the server's
configuration, error, and log files are kept.
#
Do not add a slash at the end of the directory path. If you point
ServerRoot at a non-local disk, be sure to specify a local disk on the
Mutex directive, if file-based mutexes are used. If you wish to share the
same ServerRoot for multiple httpd daemons, you will need to change at
least PidFile.
#
Define SRVROOT "C:/Apache24/Apache24"
ServerRoot "${SRVROOT}"
#
Mutex: Allows you to set the mutex mechanism and mutex file directory
for individual mutexes, or change the global defaults
#
Uncomment and change the directory if mutexes are file-based and the default
mutex file directory is not on a local disk or is not appropriate for some
other reason.
#
Mutex default:logs
Restore
Type here to search
31°C
ENG
US
14:29
23-02-2026

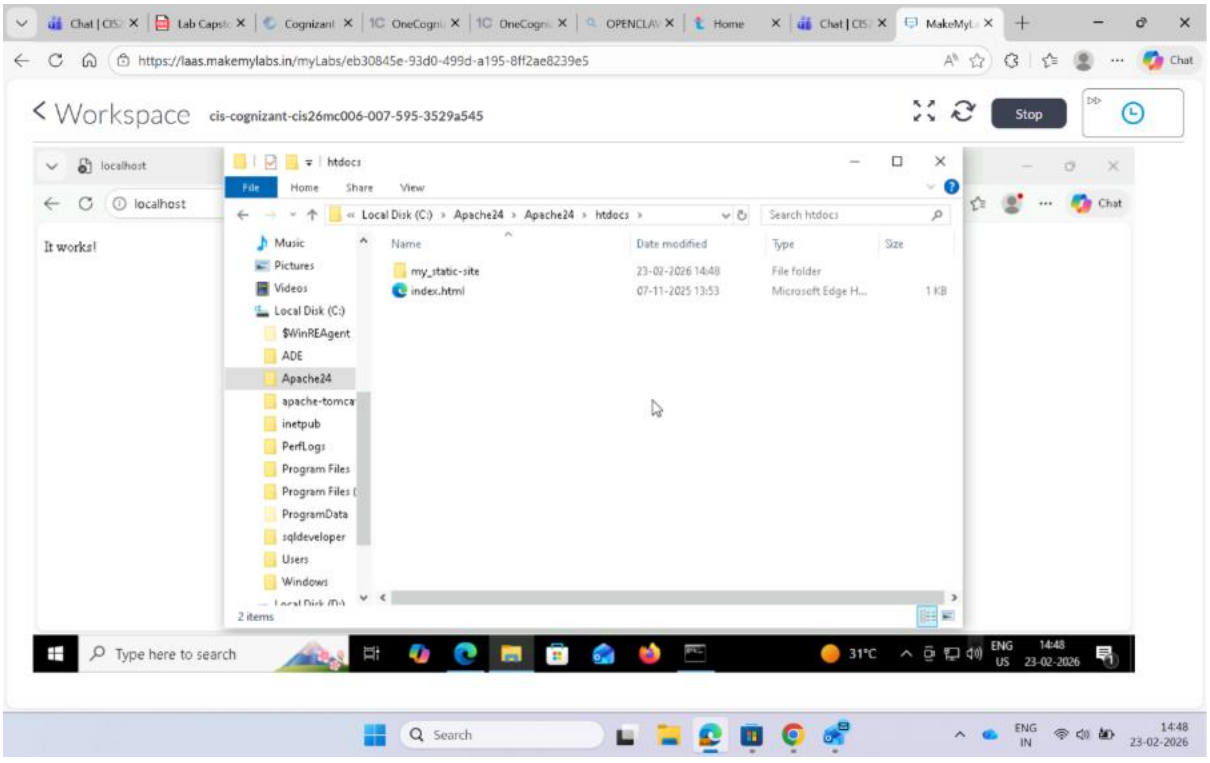

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
C:\Apache24\Apache24\conf\original\httpd.conf - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
httpd.conf
#Listen 12.34.56.78:80
Listen 80
Dynamic Shared Object (DSO) Support
To be able to use the functionality of a module which was built as a DSO you
have to place corresponding `LoadModule' lines at this location so the
directives contained in it are actually available _before_ they are used.
Statically compiled modules (those listed by `httpd -l') do not need
to be loaded here.
Example:
LoadModule foo_module modules/mod_foo.so
#LoadModule access_compat_module modules/mod_access_compat.so
LoadModule allowmethods_module modules/mod_allowmethods.so
LoadModule auth_basic_module modules/mod_auth_basic.so
LoadModule auth_digest_module modules/mod_auth_digest.so
LoadModule auth_form_module modules/mod_auth_form.so
LoadModule authn_anon_module modules/mod_authn_anon.so
LoadModule authn_core_module modules/mod_authn_core.so
LoadModule authn_dbd_module modules/mod_authn_dbd.so
Restore
Chat
Type here to search
31°C
ENG US
14:29
23-02-2026
Search
ENG IN
14:29
23-02-2026

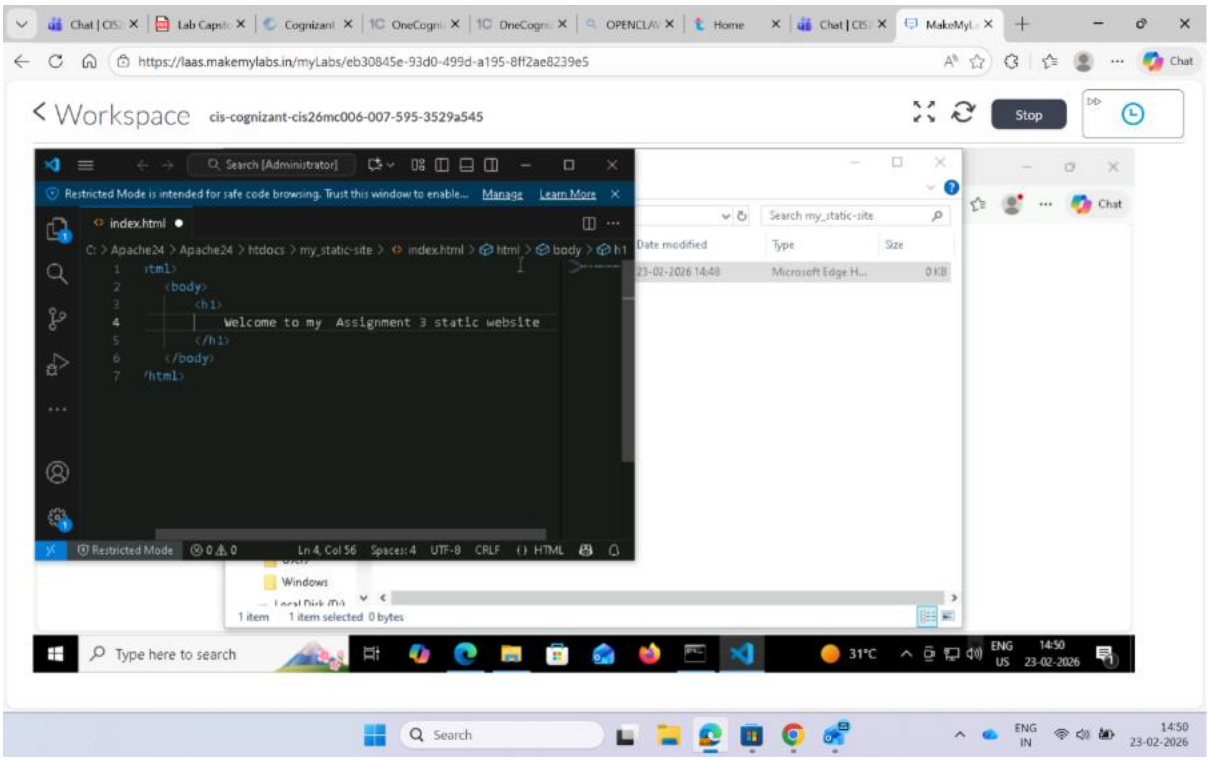

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\Apache24\Apache24\bin
C:\Apache24\Apache24\bin>httpd.exe -t
Syntax OK
C:\Apache24\Apache24\bin>httpd.exe -k start
C:\Apache24\Apache24\bin>
Chat
Type here to search
31°C
ENG US
14:44
23-02-2026
Search
ENG IN
14:44
23-02-2026

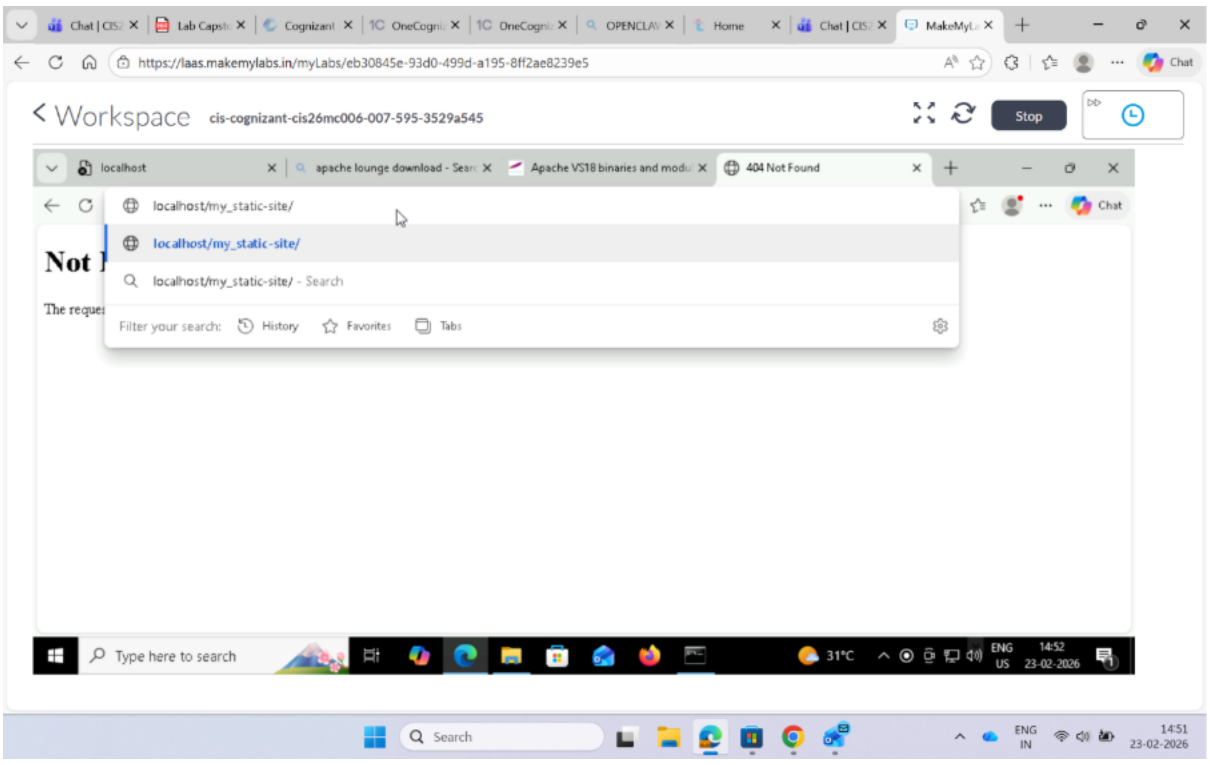

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\Apache24\Apache24\bin
C:\Apache24\Apache24\bin>httpd.exe -t
Syntax OK
C:\Apache24\Apache24\bin>httpd.exe -k start
C:\Apache24\Apache24\bin>httpd.exe -k install
[Mon Feb 23 14:45:15.717472 2026] [mpm_winnt:error] [pid 5420:tid 424] AH00433: Apache2.4: Service is already installed.
C:\Apache24\Apache24\bin>
Type here to search
31°C
ENG
US
14:45
23-02-2026

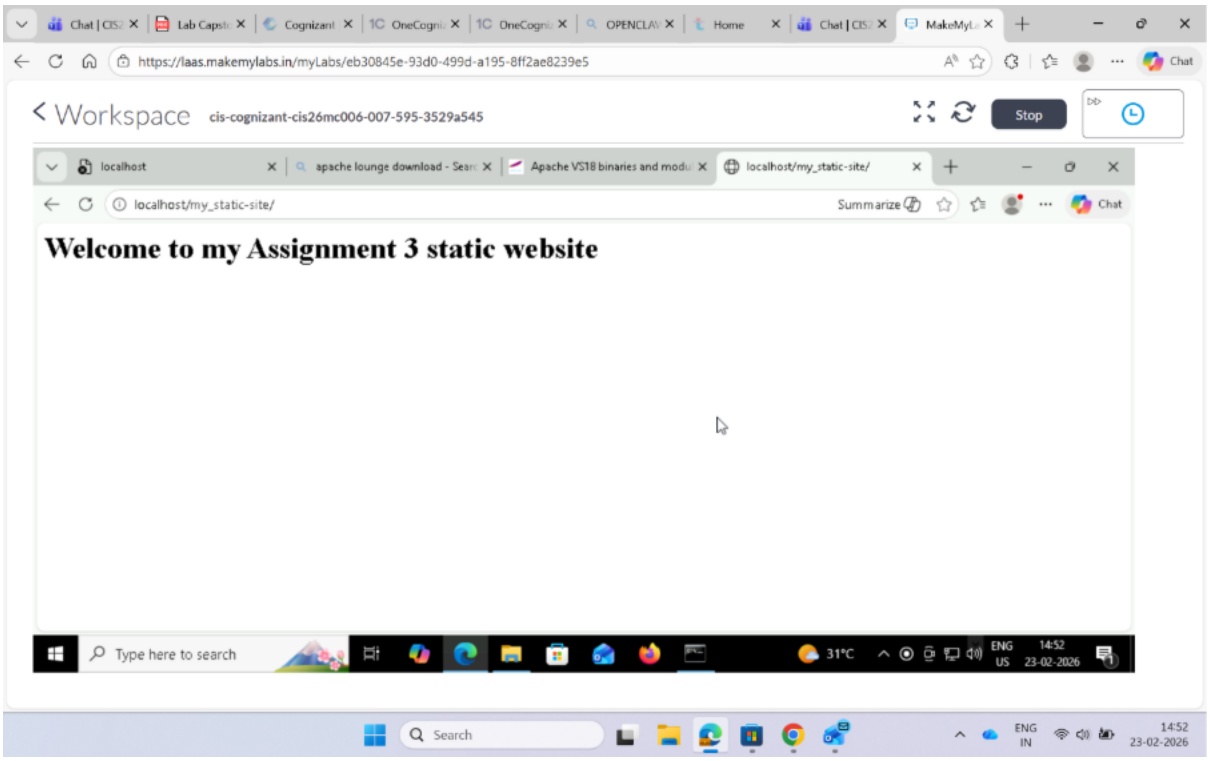

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>cd C:\Apache24\Apache24\bin
C:\Apache24\Apache24\bin>httpd.exe -t
Syntax OK
C:\Apache24\Apache24\bin>httpd.exe -k start
C:\Apache24\Apache24\bin>httpd.exe -k install
[Mon Feb 23 14:45:15.717472 2026] [mpm_winnt:error] [pid 5420:tid 424] AH00433: Apache2.4: Service is already installed.
C:\Apache24\Apache24\bin>httpd.exe -k start
(OS 10048)Only one usage of each socket address (protocol/network address/port) is normally permitted. : AH00072: make_
sock: could not bind to address [::]:80
(OS 10048)Only one usage of each socket address (protocol/network address/port) is normally permitted. : AH00072: make_
sock: could not bind to address 0.0.0.0:80
AH00451: no listening sockets available, shutting down
AH00015: Unable to open logs
C:\Apache24\Apache24\bin>
Type here to search
31°C
ENG
US
14:45
23-02-2026

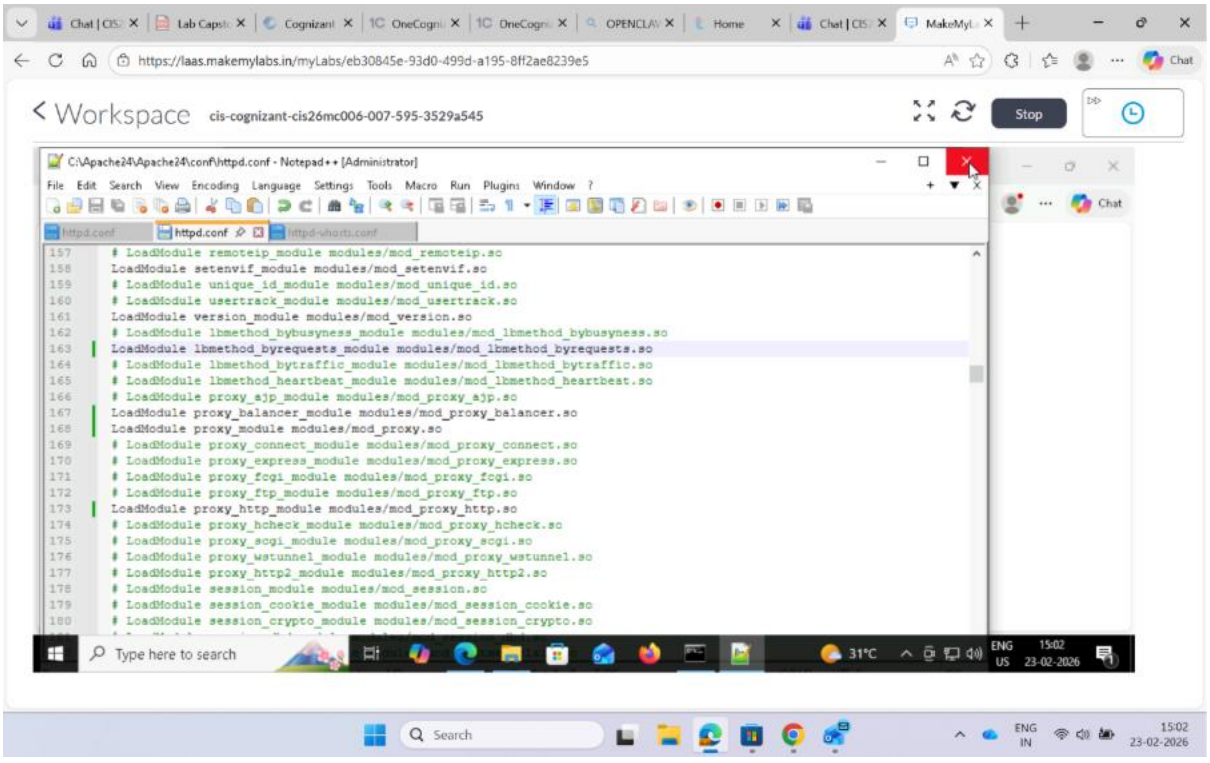

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
localhost
It works!
Type here to search
31°C
ENG
US
14:45
23-02-2026

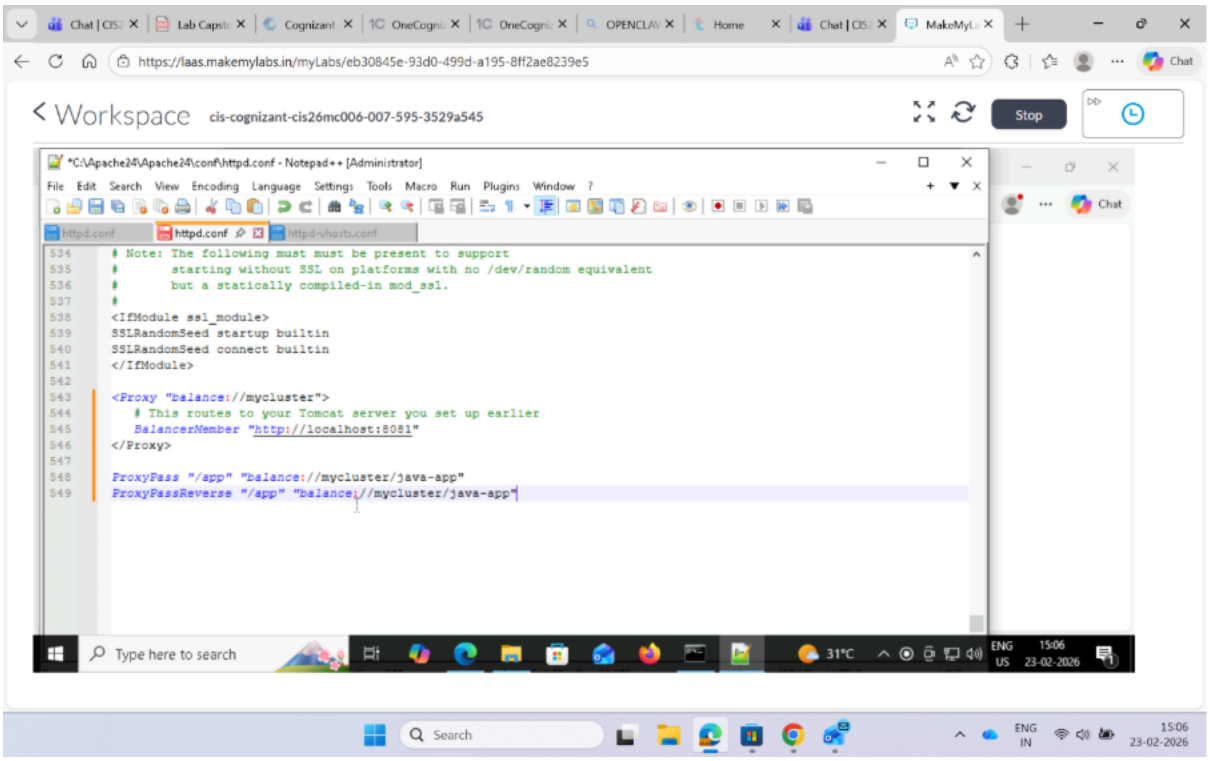

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
localhost
It works!
htdocs
Local Disk (C:) › Apache24 › Apache24 › htdocs
my_static-site
index.html
Apache24
Type here to search
31°C
ENG
US
14:48
23-02-2026

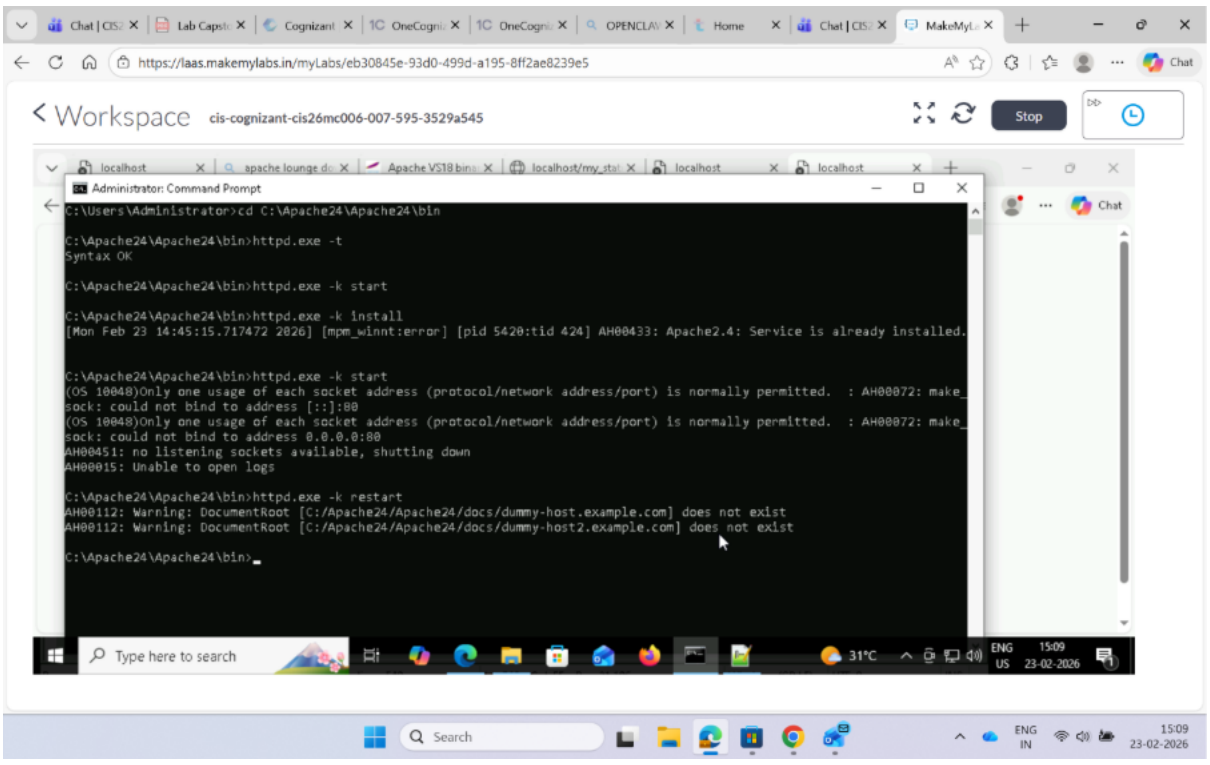

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
index.html
<html>
<body>
<h1>
Welcome to my Assignment 3 static website
</h1>
</body>
</html>
Restricted Mode is intended for safe code browsing. Trust this window to enable...
Manage
Learn More
Search my_static-site
Date modified
Type
Size
Windows
1 item
1 item selected 0 bytes
Type here to search
31°C
ENG
US
14:50
23-02-2026

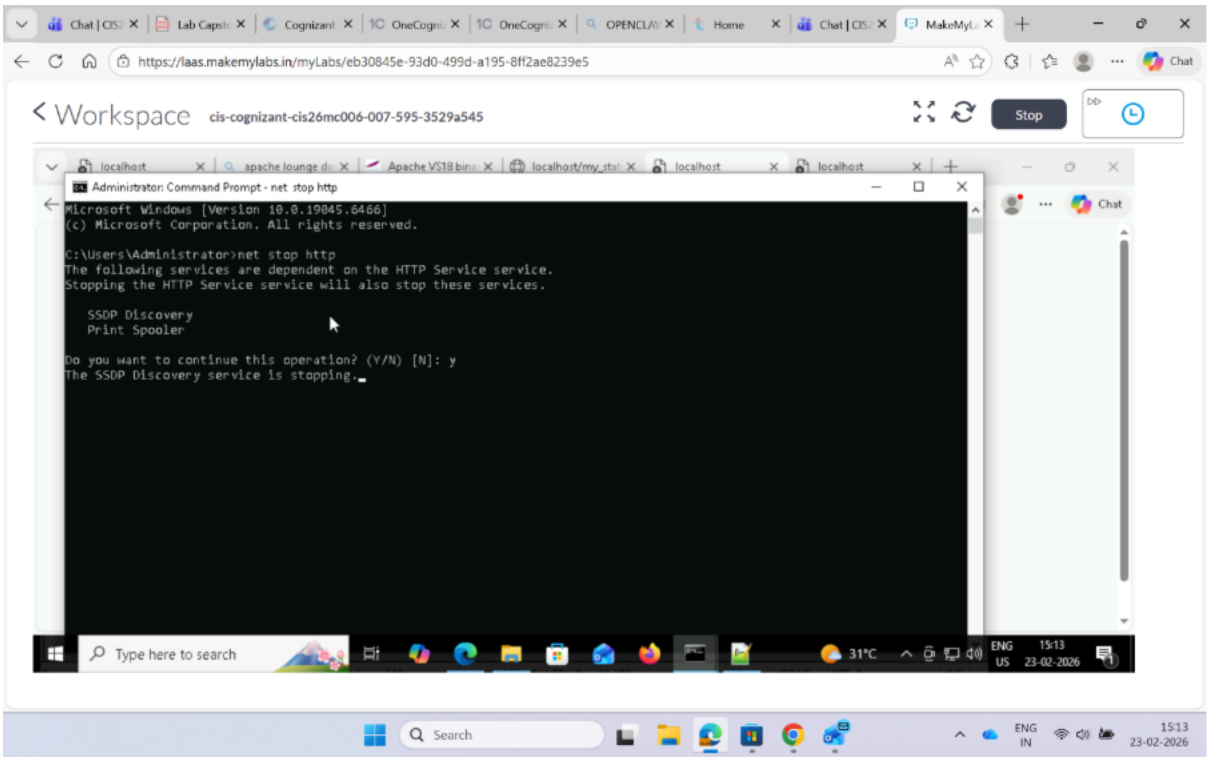

Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
localhost
404 Not Found
localhost/my_static-site/
localhost/my_static-site/
localhost/my_static-site/ - Search
Filter your search:
History
Favorites
Tabs
Not
The reque
Type here to search
31°C
ENG
US
14:52
23-02-2026

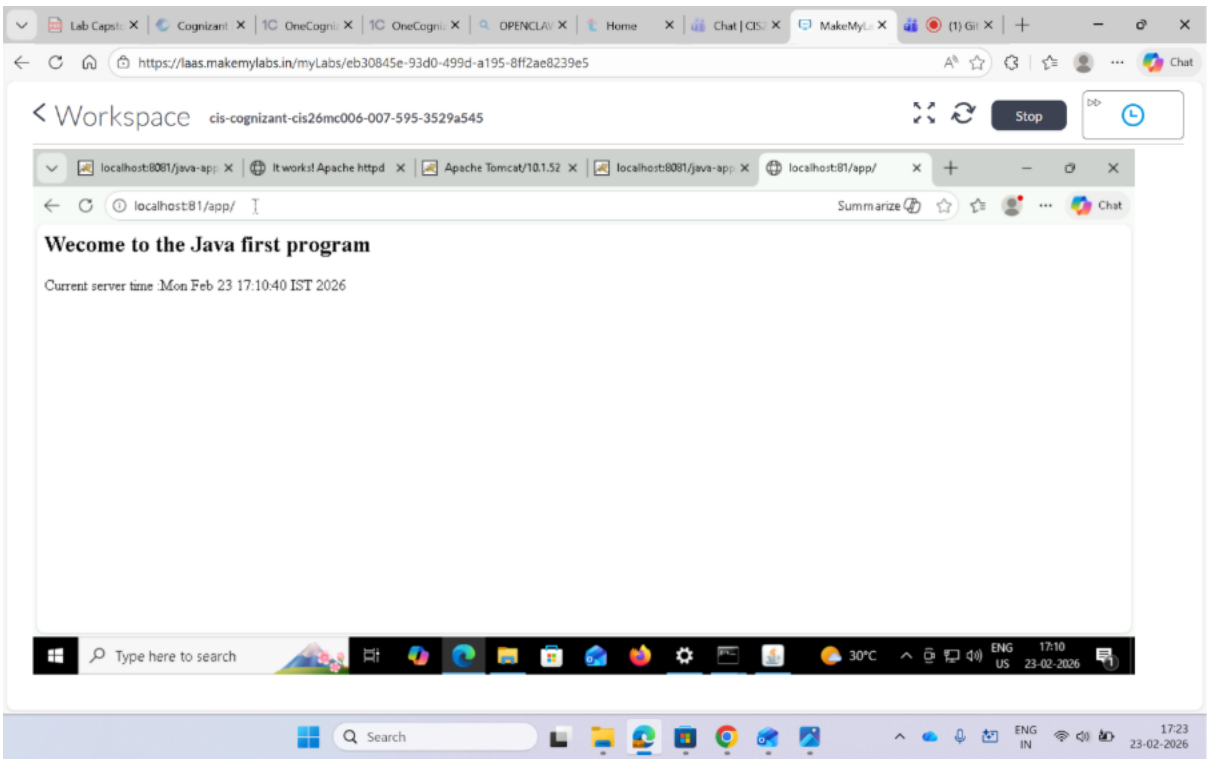

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
localhost/my_static_site/
Welcome to my Assignment 3 static website
Type here to search
31°C
ENG US
14:52
23-02-2026

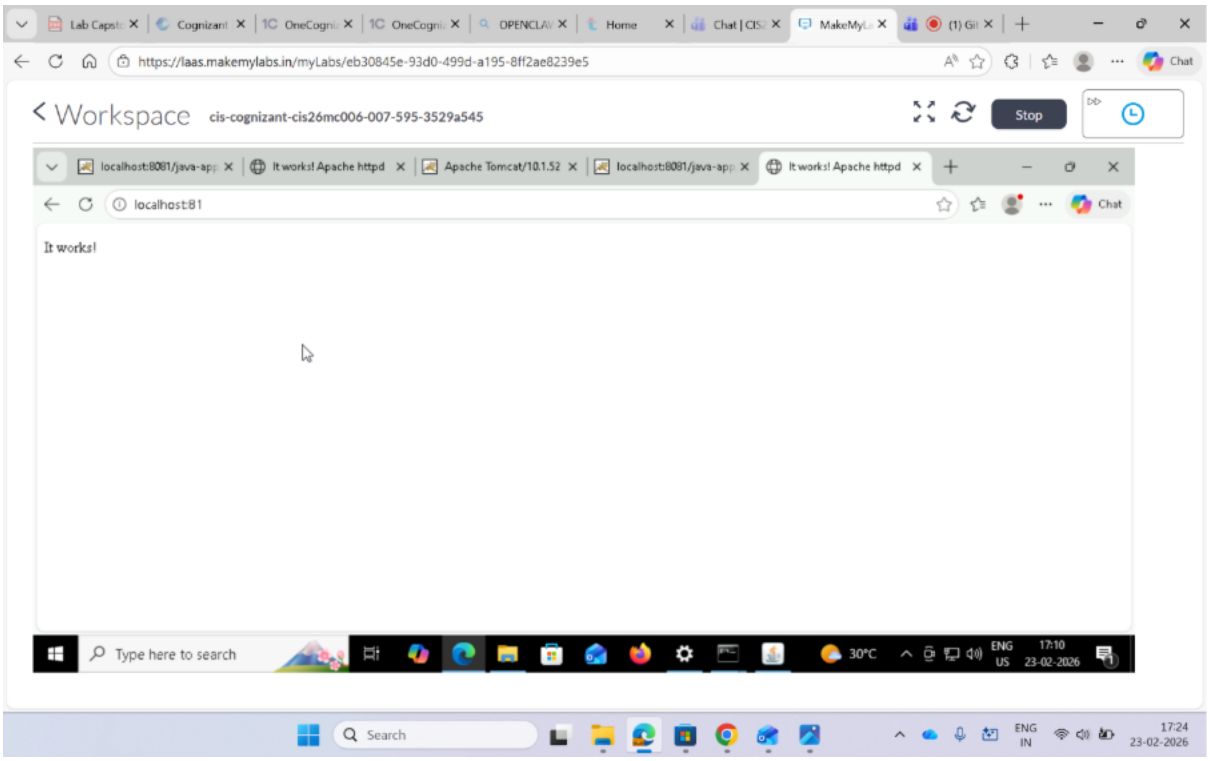

https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
C:\Apache24\Apache24\conf\httpd.conf - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
httpd.conf
157 # LoadModule remoteip_module modules/mod_remoteip.so
158 LoadModule setenvif_module modules/mod_setenvif.so
159 # LoadModule unique_id_module modules/mod_unique_id.so
160 # LoadModule usertrack_module modules/mod_usertrack.so
161 LoadModule version_module modules/mod_version.so
162 # LoadModule lbmethod_bybusyness_module modules/mod_lbmethod_bybusyness.so
163 LoadModule lbmethod_byrequests_module modules/mod_lbmethod_byrequests.so
164 # LoadModule lbmethod_bytraffic_module modules/mod_lbmethod_bytraffic.so
165 # LoadModule lbmethod_heartbeat_module modules/mod_lbmethod_heartbeat.so
166 # LoadModule proxy_ajp_module modules/mod_proxy_ajp.so
167 LoadModule proxy_balancer_module modules/mod_proxy_balancer.so
168 LoadModule proxy_module modules/mod_proxy.so
169 # LoadModule proxy_connect_module modules/mod_proxy_connect.so
170 # LoadModule proxy_express_module modules/mod_proxy_express.so
171 # LoadModule proxy_fcgi_module modules/mod_proxy_fcgi.so
172 # LoadModule proxy_ftp_module modules/mod_proxy_ftp.so
173 LoadModule proxy_http_module modules/mod_proxy_http.so
174 # LoadModule proxy_hcheck_module modules/mod_proxy_hcheck.so
175 # LoadModule proxy_scgi_module modules/mod_proxy_scgi.so
176 # LoadModule proxy_wstunnel_module modules/mod_proxy_wstunnel.so
177 # LoadModule proxy_http2_module modules/mod_proxy_http2.so
178 # LoadModule session_module modules/mod_session.so
179 # LoadModule session_cookie_module modules/mod_session_cookie.so
180 # LoadModule session_crypto_module modules/mod_session_crypto.so
Type here to search
31°C
ENG US
15:02
23-02-2026


https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
*C:\Apache24\Apache24\conf\httpd.conf - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
httpd.conf
httpd-vhosts.conf
534 # Note: The following must must be present to support
535 # starting without SSL on platforms with no /dev/random equivalent
536 # but a statically compiled-in mod_ssl.
537 #
538 <IfModule ssl_module>
539 SSLRandomSeed startup builtin
540 SSLRandomSeed connect builtin
541 </IfModule>
542
543 <Proxy "balance://mycluster">
544 # This routes to your Tomcat server you set up earlier
545 BalancerMember "http://localhost:8081"
546 </Proxy>
547
548 ProxyPass "/app" "balance://mycluster/java-app"
549 ProxyPassReverse "/app" "balance://mycluster/java-app"
Type here to search
31°C
ENG US
15:06
23-02-2026
Search
ENG IN
15:06
23-02-2026


https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Administrator: Command Prompt
C:\Users\Administrator>cd C:\Apache24\Apache24\bin
C:\Apache24\Apache24\bin>httpd.exe -t
Syntax OK
C:\Apache24\Apache24\bin>httpd.exe -k start
C:\Apache24\Apache24\bin>httpd.exe -k install
[Mon Feb 23 14:45:15.717472 2026] [mpm_winnt:error] [pid 5420:tid 424] AH00433: Apache2.4: Service is already installed.
C:\Apache24\Apache24\bin>httpd.exe -k start
(OS 10048)Only one usage of each socket address (protocol/network address/port) is normally permitted. : AH00072: make_
sock: could not bind to address [::]:80
(OS 10048)Only one usage of each socket address (protocol/network address/port) is normally permitted. : AH00072: make_
sock: could not bind to address 0.0.0.0:80
AH00451: no listening sockets available, shutting down
AH00015: Unable to open logs
C:\Apache24\Apache24\bin>httpd.exe -k restart
AH00112: Warning: DocumentRoot [C:/Apache24/Apache24/docs/dummy-host.example.com] does not exist
AH00112: Warning: DocumentRoot [C:/Apache24/Apache24/docs/dummy-host2.example.com] does not exist
C:\Apache24\Apache24\bin>
Type here to search
31°C
ENG US
15:09
23-02-2026
Search
ENG IN
15:09
23-02-2026


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
Administrator: Command Prompt - net stop http
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator>net stop http
The following services are dependent on the HTTP Service service.
Stopping the HTTP Service service will also stop these services.
SSDP Discovery
Print Spooler
Do you want to continue this operation? (Y/N) [N]: y
The SSDP Discovery service is stopping.
Type here to search
31°C
ENG
US
15:13
23-02-2026


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
localhost:81/app/
Wecome to the Java first program
Current server time :Mon Feb 23 17:10:40 IST 2026
Type here to search
30°C
ENG
US
17:10
23-02-2026


Workspace
cis-cognizant-cis26mc006-007-595-3529a545
Stop
https://laas.makemylabs.in/myLabs/eb30845e-93d0-499d-a195-8ff2ae8239e5
Chat
localhost:81
It works!
Type here to search
30°C
ENG
US
17:10
23-02-2026
Search
ENG
IN
17:24
23-02-2026